



Microsoft Surface Laptop 7th Edition for Business Service Guide

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This document and the information it contains are subject to change without notice. You can find the latest information on Surface device servicing and repair at <https://aka.ms/surfaceservicing> . Always consult the most up-to-date information available before performing device service or repair.

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Introduction

This Service Guide provides instructions for repairing Microsoft Surface devices using genuine Microsoft parts. It is intended for technically inclined individuals with the knowledge, experience, and specialized tools required to repair Microsoft devices.

IMPORTANT: Read this Guide in its entirety before starting any repairs. If at any point you are unsure or uncomfortable about performing the repairs, as detailed in this Guide, **DO NOT** proceed. Contact Microsoft for additional support options.

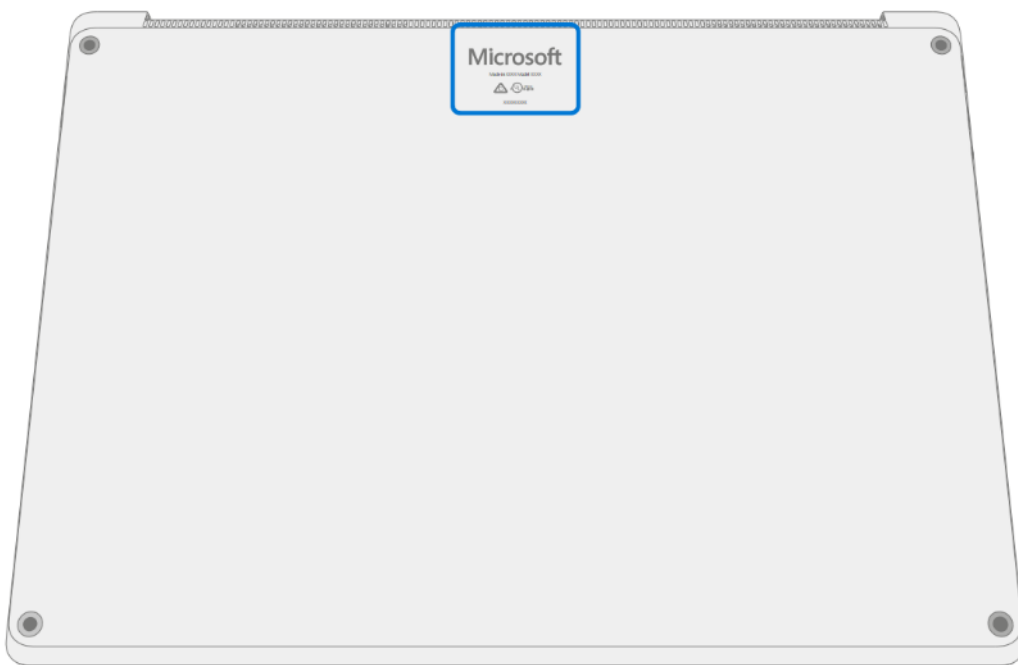
⚠ WARNING: Failure to follow the instructions in this Guide; use of non-Microsoft (non-genuine), incompatible, or modified replacement parts; and/or failure to use proper tools could result in serious personal injury, death, and/or damage to the product or other property.

Device Identity Information

- Surface Laptop 7th Edition

Support Link – [Link](#)

The model and serial number for Surface Laptops is on the bottom center closest to the display hinge point.



General Information, Precautions, and Warnings

Tools

This section documents the tools recommended or required by Microsoft to successfully complete a repair on a Surface device. Microsoft Service Tools (recommended and required) are sold by iFixit in partnership with Microsoft. Items under Electronic Repair Hardware and Tools can be commonly purchased from electronic repair retailers. Lastly, items under standard tools and misc. items on this list can be commonly purchased from consumer retailers.

Recommended Microsoft Service Tools

ESD-safe Surface Battery Cover - iFixit	
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Required Microsoft Service Tools

Required Electronic Repair Hardware or Tools

Anti-static Wrist Strap (1 MOhm resistance)	
ESD-safe mat or benchtop	
Nylon Spudger/Probing Tool	
Plastic Opening Pick	

Plastic Opening Tool	
ESD-safe Tweezers	

Required Standard Tools and Misc Items

- 2IP Torx-Plus Driver
- 3IP Torx-Plus Driver
- 5IP Torx-Plus Driver
- 6IP Torx-Plus Driver
- USB 3.0 Thumb drive – 16 GB minimum storage
- Isopropyl alcohol dispenser bottle (use 70% IPA)
- Cleaning swabs
- Microfiber Cloth
- Lint free cleaning cloth
- 4 Gallon Bucket
- 2.0 Gallons Sand, Clean
- 65W Microsoft Surface Power Supply
- 0.1mm Thickness Gauge
- 0.15mm Thickness Gauge

General Safety Precautions

 Always observe the following general safety precautions:

- Opening and/or repairing any electronic device can present a risk of electric shock, fire, serious personal injury, death, damage to the device or other property, and/or other hazards. Exercise caution when undertaking the repair activities described in this Guide. The repair activities identified in this Guide should only be undertaken by technically inclined individuals with the knowledge, experience, and specialized tools required to repair Microsoft devices.
- Improper use or handling of devices or their batteries may result in fire or explosion. Only open the enclosure on a device as outlined in this Guide.
- Do not heat, puncture, mutilate, or dispose of devices or their batteries in fire. Do not leave or charge devices in direct sunlight or expose devices or their batteries to temperatures outside the recommended operating range of 0°C to 60°C/32°F to 140°F for an extended period. Doing so can result in battery failure, electric shock, fire, serious personal injury, death, and/or damage to the device or other property.
- We recommend wearing protective eyewear and gloves when disassembling/re-assembling a device.
- Clean your work surface regularly to remove debris and abrasive particles.
- While working on devices, avoid the use of clothing accessories such as bracelets, rings, or watches that can cause electrical shorts and/or damage the battery.
- As you remove each subassembly from the device, place the subassembly (and all accompanying screws) away from the work area to prevent damage to the device or to the subassembly.
- If battery damage (e.g., leaking, expansion, folds or other) is discovered during device repair or if the battery is impacted or damaged during replacement, **DO NOT** proceed. Refer to the [Actions to take in case of a Thermal Event](#) section or contact Microsoft directly for proper device disposition.

For additional product safety information relevant to Microsoft Surface devices, see aka.ms/surface-safety or the Surface app. To open the Surface app, select the Start button, enter Surface into the search box, then select the Surface app.

Electro-Static Discharge (ESD) Prevention

- Review and follow the general guidelines and ESD prevention steps in this Guide prior to beginning work.
- Ensure your work surface is level/flat and covered with ESD-safe, soft, non-marring material.
- Before opening a device, always wear an anti-static wrist strap and confirm your work area is properly grounded to protect vulnerable electronics from electrostatic discharge (ESD).
- Parts removed from a device during the repair process should be stored in ESD-safe bags and packaged for return or recycling in the same packaging that the new replacement part came in.

Repair-Specific Precautions and Warnings

- For Autopilot managed Surface Products refer to the following guidelines posted [here](#).

⚠ WARNING: Before opening a device, ensure it is powered off and disconnected from its power source. Disconnect the device charger or power cord from mains power.

- For devices with rechargeable lithium-ion batteries that power on, fully discharge the battery before beginning repair. To expedite the battery discharge process:
 - Disconnect the charger from the device.
 - Increase display brightness to the highest level.
 - Turn on wi-fi and Bluetooth.
 - Open the Camera app in Windows.
 - Play music or video files from a local drive or streaming service.
- Operate the device in this mode until the battery is fully discharged and the device powers off.
- Review the General Safety Precautions and Battery Safety Sections of this Guide before beginning work.

⚠ WARNING: For Surface devices where the battery is affixed to the back cover, place the back cover with the battery in a location where it will be protected from possible punctures, impacts, crushing, or drops during the repair process. Refer to the [Battery Safety](#) section in this guide for more information.

⚠ WARNING: During all activities (excluding feet-only replacement) check to ensure that no loose articles are on the back cover or remain inside the device before reassembling it.

IMPORTANT: Remove the rSSD (removable Solid-State Drive) whenever the Keyboard is removed from the device. rSSD removal disconnects the battery from all device logical components for safety purposes. Refer to Procedure-Removal (rSSD) section for details.

IMPORTANT: Whenever the rSSD has been removed, powering on the device requires that the rSSD and Keyboard are installed.

IMPORTANT: The serial number for this device model is located on its original cover. To keep track of the device's serial number, please record it using waterproof ink on a sticker or label and apply the sticker or label to an easily accessible area on the device exterior. For serial number location please see the [Device Identity Information](#) section. The serial number cannot be added permanently to a replacement part. Microsoft may have provided a label for this use in the replacement part's packaging

Battery Safety

- This device contains a built-in, lithium-ion rechargeable battery. Battery safety is a significant concern when repairing a device.
- For optimum compatibility, performance, and product safety, we recommend using genuine Microsoft replacement parts available on [Microsoft.com](https://www.microsoft.com) and other online part retailers such as iFixit. Use of non-Microsoft (non-genuine), incompatible, reused, or modified batteries; improper battery installation; improper handling or storage of batteries; and/or failure to follow the instructions in this Guide could cause battery overheating, expansion, venting, leaking, or a thermal event which could result in fire, serious personal injury, death, data loss, or damage to the device or other property damage.
- Before beginning device repair, ensure your workspace is free of flammable debris or materials, has adequate ventilation, and that you have a fire suppressant device (example: fire blanket, container of sand, Class B fire extinguisher) within easy reach or you are within 20 feet of a fireproof enclosure. Fireproof enclosures should be kept free of combustible or flammable materials.

WARNING

It is recommended that an ESD-safe battery cover be placed across the device to protect the battery from any physical contact or accidental damage whenever the display is removed for internal repairs. Ensure corners of cover are always aligned with the corners of the device while battery is exposed. If the battery cover is misaligned during activities in any way, re-align before continuing work.

- Use personal protective equipment (PPE) when handling damaged, venting, or hot battery packs.
- Use the following best practices when handling batteries:
 - Always fully discharge batteries by running an application such as video playback with the device unplugged. If the device does not function while unplugged, you may leave out this step.
 - Do not puncture, impact, strike, bend, or crush the battery or a device containing a battery.
 - Keep your workspace clear of debris, extra tools, and sharp objects.
 - Exercise caution when using sharp tools near the battery to avoid impacting or poking the battery.
 - Do not leave loose screws or small parts inside the device.
 - Avoid using tools that conduct electricity.
 - Do not drop or throw a lithium-ion battery.
 - Do not expose the battery to excessive heat, sunlight, or temperatures outside the battery's normal operating range (0°C to 60°C) / (32°F to 140°F)
 - Ensure you handle, recycle, and/or dispose of used or damaged batteries in accordance with local laws and regulations. Follow Handling Used, Damaged or Defective Li-ion Batteries below.
- If the device repair cannot be completed immediately and the device needs to be stored temporarily before restarting the repair
 - Select a storage location and process that follows the battery safety precautions in this Guide.
 - Avoid exposing the device to environmental conditions and objects that could damage the battery pack.

- Reinspect the battery pack as outlined in this Guide prior to restarting repair and installing the new battery pack.

Battery Warning Level

⚠ WARNING: Please note that the battery bears the following warning label. Please heed the information provided on the label.



Battery is replaceable by trained personnel; replacement must follow Microsoft procedure
See <http://aka.ms/surface-safety> for important information.

- Risk of fire or burning – contact Microsoft for assistance
- Do not modify battery, its wiring, or connectors
- Do not short circuit, bend, dent, crush, or puncture battery
- Do not dispose of battery in fire or expose to high temperatures (+140°F/60°C)



Lithium-Ion Battery Inspection

Upon device opening, we recommend that you visually inspect the battery for signs of damage. Factors to consider when inspecting the battery include, but are not limited to:

- Evidence of leaking or venting
- Visible signs of physical or mechanical damage, such as:
 - Expansion or swelling. In expanded or swollen batteries, the soft pouch encasing the cell pulls away from the inner material and appears baggy, loose, or puffy.
 - Discoloration of the battery casing.
 - Odor, smell, or visible corrosion. Leaked battery electrolyte smells like nail polish remover (acetone).
 - Dents along the battery cell edges or on the top surface.
 - Surface scratches that have exposed the aluminum beneath the black coating layer on the battery.
 - Loose or damaged wires.
 - Known misuse or abuse.

Any battery exhibiting the signs listed above must be replaced. Consult the [Battery Replacement Process](#) section of this document for battery replacement instructions.

Handling Used, Damaged, or Defective Lithium-Ion Batteries

- **DO NOT** dispose of used lithium-ion batteries, whether damaged or not, in household or commercial garbage or recycling bins.



WARNING: DO NOT SHIP DAMAGED OR DEFECTIVE BATTERIES ALONE OR INSIDE DEVICES.

Damaged or defective batteries and devices containing damaged or defective batteries require special packaging and handling.

Prior to transport:

- Follow all instructions provided by your local e-waste recycling or household hazardous waste collection provider.
- Place the device or battery in individual, non-metallic inner packaging, such as a zip-to-close plastic bag, that completely encloses the device or battery.
- Surround the inner packaging with non-combustible, electrically non-conductive, absorbent cushioning material.
- Each damaged battery or device containing a damaged battery should be packed individually in its own carton and that carton should be clearly marked as containing a damaged battery.

For more information on industry practices concerning damaged, defective, or recalled batteries, please see [PHMSA Lithium-Battery-Recycling-Safety-Advisory](#).

Undamaged, used lithium batteries can be sent to e-waste recycling or household hazardous waste collection points for processing. Please see <https://www.microsoft.com/en-us/legal/compliance/recycling> for more information.

Actions to take in case of a Thermal Event

- **DO NOT** use water. Immediately smother the battery or device with clean, dry sand, a fire blanket, or an appropriate (Class B) fire extinguisher. If using sand, dump the sand all at once until the device is completely covered.
- Contact local fire authorities if further assistance is needed.
- Exit the work area and ventilate it until it is clear of smoke.
- Wait at least 2 hours before attempting to touch the device.
- Dispose of the damaged battery or device in accordance with local environmental or e-waste laws and guidelines.

Report Battery Thermal Events to Microsoft

A thermal event is a rapid chemical chain reaction that can occur inside a battery cell. During a thermal event, the energy stored inside the battery is released suddenly, resulting in heating and/or smoke and, in some instances, fire or flame. A battery thermal event can be triggered by physical damage to the battery (including during replacement/repair), improper storage, or exposure to temperatures outside of the battery's operating range.

Act immediately if you see any of the following symptoms of a battery thermal event:

- Smoke, soot, sparks, or flame emitted by the battery or from a device containing a battery.
- The battery pouch suddenly expands in size.
- A popping or hissing noise from the battery or a device containing a battery.

Stop Repair and Contact Microsoft

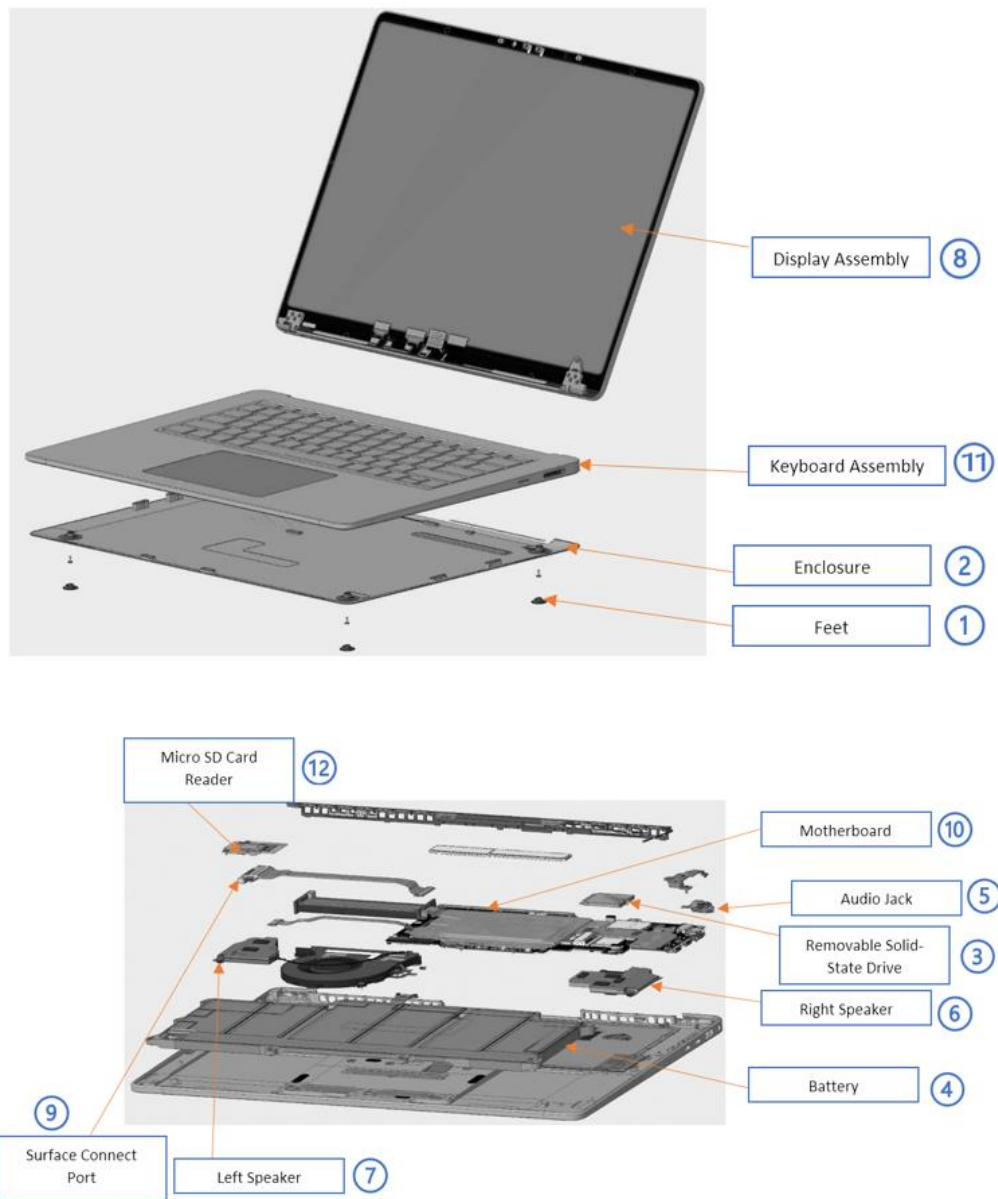
If any Microsoft device visually exhibits any of the following symptoms, cease all further repair efforts and contact Microsoft Surface Customer Support to report and obtain next steps:

- Any burned or melted components, traces, or plastic parts on the outside of the device, or which otherwise exhibits heat damage, including charring seen in charging and other ports.
- Any burned or melted components, traces, or plastic parts on the inside of the device, or which otherwise exhibits heat damage.
- Any accessories exhibiting melting or heat damage that are included with the Microsoft device, such as power supplies, keyboards, mice, cables, charging connectors, etc.
- Any devices that exhibit a case that has separated apart or opened for reasons other than impact damage from dropping, evidence of tampering, or separation caused by a malfunctioning battery.
- Any other finding that may constitute a safety hazard to the user, such as sharp edges on plastics.

Microsoft Surface Customer Support will ask you to provide the following information:

- The model and serial number of the affected Microsoft Surface device and/or accessory(ies).
- A brief description of the damage found.
- Clear photographs depicting the symptoms observed.

Illustrated Service Parts List



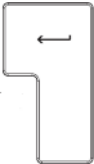
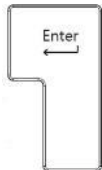
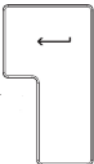
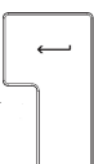
IMPORTANT: Repair workflows may require multiple parts to be ordered to complete the repair successfully. Please check the primary and additional components section in each repair workflow to ensure you have all required parts before beginning your repair.

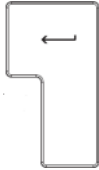
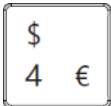
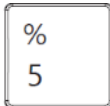
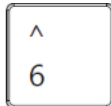
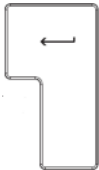
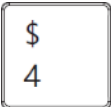
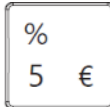
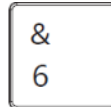
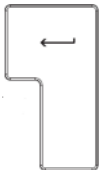
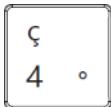
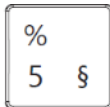
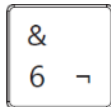
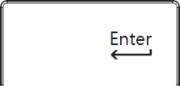
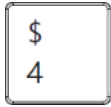
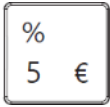
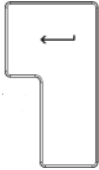
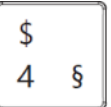
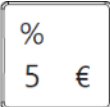
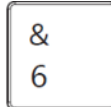
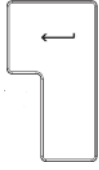
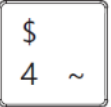
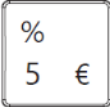
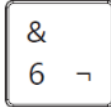
Item	Component	SKU Part No.
1	Feet	
	Platinum	EP2-29864
	Black	EP2-29865
2	Enclosure	
	Platinum – 13”	EP2-29765
	Black – 13”	EP2-29766
	Platinum – 15”	EP2-29767
	Black – 15”	EP2-29768
3	Removable Solid-State Drive	
	256GB	EP2-29866
	512GB	EP2-29867
	1TB	EP2-29868
4	Battery	
	Battery – 13”	EP2-29762
	Battery – 15”	EP2-29763
5	Audio Jack	
	Platinum	EP2-29761
	Black	EP2-29760
6	Right Speaker	
	Right Speaker – 13”	EP2-29870
	Right Speaker – 15”	EP2-29872
7	Left Speaker	
	Left Speaker – 13”	EP2-29869
	Left Speaker – 15”	EP2-29871
8	Display Assembly (Includes Camera)	
	Platinum – 13”	EP2-29746
	Black – 13”	EP2-29747
	Platinum – 15”	EP2-29753
	Black – 15”	EP2-29752
9	Surface Connect Port	
	Surface Connect Port - 13”	EP2-29769
	Surface Connect Port - 15”	EP2-29770
10	Fan	
	Fan - 13”	EP2-29771
	Fan- 15”	EP2-29772
11	Motherboard (includes main processor and main memory, and thermal module)	
	Ultra 5 16GB – 13”	EP2-29720
	Ultra 5 32GB – 13”	EP2-29721
	Ultra 7 16GB – 13”	EP2-29722
	Ultra 7 32GB – 13”	EP2-29723
	Ultra 5 16GB – 15”	EP2-29724
	Ultra 5 16GB w/ Smart Card Reader – 15”	EP2-29728
	Ultra 5 32GB – 15”	EP2-29725
	Ultra 5 32GB w/ Smart Card Reader – 15”	EP2-29729

	Ultra 7 16GB – 15”	EP2-29726
	Ultra 7 16GB w/ Smart Card Reader – 15”	EP2-29730
	Ultra 7 32GB – 15”	EP2-29727
	Ultra 7 32GB w/ Smart Card Reader – 15”	EP2-29731
12	Keyboard Assembly (includes Trackpad)	
	Platinum – Arabic – 13”	EP2-29601
	Platinum – Belgium – 13”	EP2-29607
	Platinum – Chinese Traditional (Taiwan) – 13”	EP2-29600
	Platinum – Canada EN/FR – 13”	EP2-29595
	Platinum – English International – 13”	EP2-29603
	Platinum – English UK – 13”	EP2-29602
	Platinum – English – 13”	EP2-29594
	Platinum – French – 13”	EP2-29605
	Platinum – German – 13”	EP2-29604
	Platinum – Italian – 13”	EP2-29608
	Platinum – Japanese – 13”	EP2-29597
	Platinum – Korean – 13”	EP2-29598
	Platinum – Nordic – 13”	EP2-29611
	Platinum – Portuguese – 13”	EP2-29609
	Platinum – Spanish (Mexico) – 13”	EP2-29596
	Platinum – Spanish (Spain) – 13”	EP2-29610
	Platinum – Swiss/Luxembourg – 13”	EP2-29606
	Platinum – Thai – 13”	EP2-29599
	Black – Arabic – 13”	EP2-29583
	Black – Belgium – 13”	EP2-29589
	Black – Chinese Traditional (Taiwan) – 13”	EP2-29582
	Black – Canada EN/FR – 13”	EP2-29577
	Black – English International – 13”	EP2-29585
	Black – English UK – 13”	EP2-29584
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	Platinum – Arabic – 15”	EP2-29637
	Platinum – Belgium – 15”	EP2-29643
	Platinum – Chinese Traditional (Taiwan) – 15”	EP2-29636
	Platinum – Canada EN/FR – 15”	EP2-29631
	Platinum – English International – 15”	EP2-29639

	Platinum – English UK – 15”	EP2-29638
	Platinum – English – 15”	EP2-29630
	Platinum – French – 15”	EP2-29641
	Platinum – German – 15”	EP2-29640
	Platinum – Italian – 15”	EP2-29644
	Platinum – Japanese – 15”	EP2-29633
	Platinum – Korean – 15”	EP2-29634
	Platinum – Nordic – 13”	EP2-29647
	Platinum – Portuguese – 15”	EP2-29645
	Platinum – Spanish (Mexico) – 15”	EP2-29632
	Platinum – Spanish (Spain) – 15”	EP2-29646
	Platinum – Swiss/Luxembourg – 15”	EP2-29642
	Platinum – Thai – 15”	EP2-29635
	Black – Arabic – 15”	EP2-29619
	Black – Belgium – 15”	EP2-29625
	Black – Chinese Traditional (Taiwan) – 15”	EP2-29618
	Black – Canada EN/FR – 15”	EP2-29613
	Black – English International – 15”	EP2-29621
	Black – English UK – 15”	EP2-29620
	Black – English – 15”	EP2-29612
	Black – French – 15”	EP2-29623
	Black – German – 15”	EP2-29622
	Black – Italian – 15”	EP2-29626
	Black – Japanese – 15”	EP2-29615
	Black – Korean – 15”	EP2-29616
	Black – Nordic – 13”	EP2-29629
	Black – Portuguese – 15”	EP2-29627
	Black – Spanish (Mexico) – 15”	EP2-29614
	Black – Spanish (Spain) – 15”	EP2-29628
	Black – Swiss/Luxembourg – 15”	EP2-29624
	Black – Thai – 15”	EP2-29617
	Black – Belgium – 15” w/ Smart Card Reader	EP2-36904
	Black – Canada EN/FR – 15” w/ Smart Card Reader	EP2-36897
	Black – English International – 15” w/ Smart Card Reader	EP2-36900
	Black – English UK – 15” w/ Smart Card Reader	EP2-36899
	Black – English – 15” w/ Smart Card Reader	EP2-36896
	Black – French – 15” w/ Smart Card Reader	EP2-36902
	Black – German – 15” w/ Smart Card Reader	EP2-36901
	Black – Italian – 15” w/ Smart Card Reader	EP2-36905
	Black – Japanese – 15” w/ Smart Card Reader	EP2-36898
	Black – Nordic – 13” w/ Smart Card Reader	EP2-36908
	Black – Portuguese – 15” w/ Smart Card Reader	EP2-36906
	Black – Spanish (Spain) – 15” w/ Smart Card Reader	EP2-36907
	Black – Swiss/Luxembourg – 15” w/ Smart Card Reader	EP2-36903
13	Micro SD Card Reader	
	Micro SD Card Reader	EP2-29875

14	Smart Card Reader	
	Smart Card Reader	EP2-29764

Description	Enter Key	"4,5,6" Keys
104 English, US		\$ 4 % 5 ^ 6
105 Canadian, Bilingual		\$ 4 ¢ % 5 ¢ ? 6 ^
109 Japan		\$ う 4 う % え 5 え & お 6 お
105 Austria/Germany		\$ 4 % 5 & 6
105 Belgium AZERTY		4 ' { 5 ([6 § ^
105 Nordic Denmark, Finland, Norway, Sweden		¤ 4 \$ % 5 & 6
105 French		4 ' { 5 ([6 -

105 English, UK Ireland		  
<u>Description</u>	<u>Enter Key</u>	<u>"4,5,6" Keys</u>
105 Italy		  
105 Switzerland, Luxembourg		  
104 English, International Netherlands		  
105 Portuguese		  
105 Spanish, European		  

Software Tools – Diagnostic, Calibration, and Troubleshooting

This section covers the software tools required to support a Surface device through problem discovery and resolution.

Genuine Microsoft Replacement Parts

- Genuine Microsoft replacement parts can be obtained directly from Microsoft on [Microsoft.com](https://www.microsoft.com).
- Genuine Microsoft replacement parts are also available on the partner sites below:

- [iFixit](#)

General Support

- For general Surface support, visit www.support.microsoft.com
- To troubleshoot device feature/function problems or learn more about Surface Laptops visit <https://support.microsoft.com/surface>.
- If you would like to learn more about Windows, visit <https://support.microsoft.com/windows>
- To learn more about the accessibility features of the Surface Laptop, go to the online user guide at aka.ms/Windows-Accessibility

Software Tools

- How To: [Update Surface device firmware and OS](#)
- How To: [Surface Tools Video](#)
- How To: [Surface Diagnostic Toolkit User Guide](#)
- Download: [Surface drivers and firmware](#)
- Download: [Surface Diagnostic Toolkit \(SDT\)](#)
- Download: [Surface Data Eraser](#)
- Download: [Surface Imaging Tools](#)

Calibration and Authentication

Specific components require additional software calibration or authentication after completing the installation of the component before the part will function to full capability. The specific steps will be called out in the pertinent repair workflows.

Impacted Parts

- **Display (TDM) –**
 - Pre-installation – requires a pre-installation repair workflow, completed in SDT, to put the device into repair mode.
 - Post-installation – requires a post-installation workflow, completed in SDT, to calibrate the display to the correct settings.
- **Battery –**
 - Pre-installation - requires a pre-installation repair workflow, completed in SDT, to put the device into repair mode.
 - Post-installation – requires a post-installation authentication workflow, completed in SDT, to authenticate the new battery as a valid Microsoft part.
- **Motherboard (PCBA) –**
 - Post-installation - requires a post-installation workflow for Display and an authentication for Battery, completed in SDT, to calibrate the display to the correct settings with the new board and ensure the battery is detected as an authentic part.

Hardware Troubleshooting Approach

Microsoft recommends the following approach for troubleshooting Surface devices:

1. Update the device to the latest Operating System and Driver/Firmware versions using Windows Update.

Important: Ensuring your device is fully up to date is important for ensuring the issue you are experiencing is not fixed by a software update prior to conducting a hardware repair.

2. Utilize the Surface Diagnostic Toolkit (SDT) after confirming the device is fully updated to confirm the hardware fault is still present prior to repair.
3. After the repair is completed, run the Surface Diagnostic Toolkit (SDT) to validate the original hardware fault is resolved.
 - a. If the issue is still being experienced, it is recommended to reimage the device using a Surface Recovery Image (BMR) to return the device to a known OS/FW state. Additional repairs should only be conducted if the issue persists after re-imaging the device.

Component Removal and Replacement Procedures

Prerequisite Steps

Steps outlined in this section should be conducted prior to starting any repair on a Surface device.

- **Power off device** – Ensure the device is powered off completely and the battery has been fully discharged. Refer to the [Repair-Specific Precautions and Warnings section](#) for details. Once discharged, the device should be disconnected from all power sources.
- **ESD Prevention** – Ensure ESD prevention steps and general guidelines are followed prior to opening the device. Refer to the [ESD Prevention section](#) for details.

Position Device – To prevent damage to the device, ensure the device is placed on a clean surface free of debris.

Feet Replacement

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

Required Tools

- Plastic Opening Pick
- Soft ESD-Safe Mat

Primary Components

- Feet (Refer to the Illustrated Service Parts List)

Additional Components (Ordered Separately)

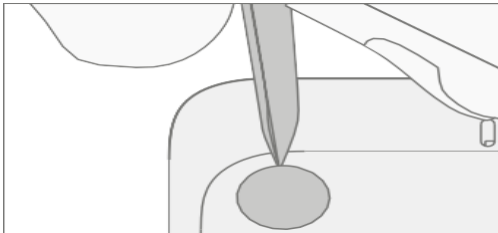
- N/A

Procedure – Removal (Feet)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.



2. **Remove the Feet** – Using the Nylon Spudger, carefully pry up all four feet to expose the screws.



Procedure – Installation (Feet)

1. **Install the Feet** - Line up the posts on the Foot to the matching hole pattern on the Enclosure. Press firmly until the foot clicks into place. Repeat for the other 3 feet.



Enclosure Replacement

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

- **Device Serial Number Notation** – The replacement Enclosure supplied for repair will not have a serial number. To ensure the customer has the best experience with future Microsoft support cases, it is recommended to create a notation of the device serial number and provide it to the customer upon completion of the repair.

Required Tools

- Plastic Opening Pick
- Soft ESD-Safe Mat
- 5IP (Torx-Plus) Driver

Primary Components

- Enclosure (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)

Additional Components (Ordered Separately)

- Feet (Refer to the Illustrated Service Parts List)

Procedure – Removal (Enclosure)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.
3. **Remove the Enclosure screws** – Using a 5IP (Torx-Plus) driver, uninstall each of the 4 screws from under the feet.



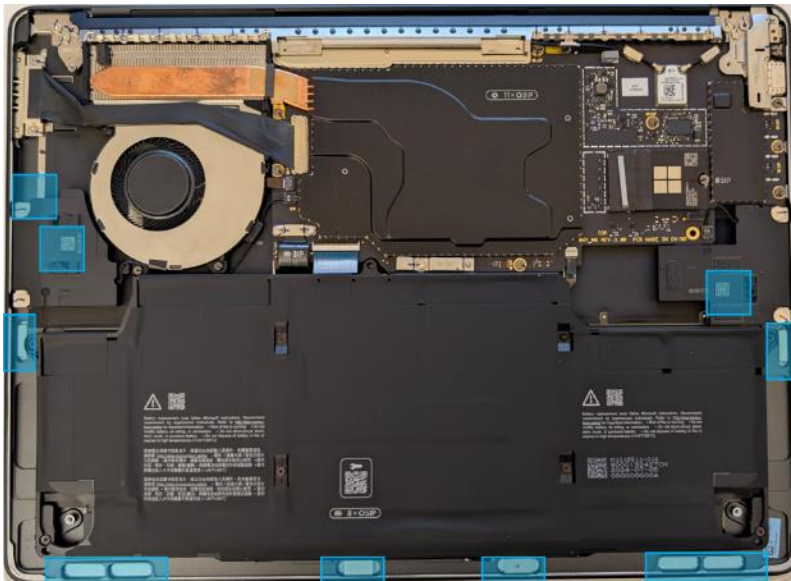
4. **Separate the Enclosure from the Device** – Carefully insert the tip of a Plastic Opening Pick into the space between the Enclosure and the Keyboard Assembly as shown below. Move the Plastic Opening Pick around the perimeter of the device to create a gap. Using both hands, carefully lift the Enclosure off the device and place it on a soft ESD-Safe mat.



Caution: Do Not lift the Enclosure from the black plastic antenna.

Procedure – Installation (Enclosure)

1. **Check for unexpected items within the Device** - Perform a thorough visual inspection of the device interior, or Enclosure if re-using, for any loose articles that may be present. Of specific importance are the magnets (identified below).



2. **Install the Enclosure** - Using both hands, carefully lower the top or bottom edge of the Enclosure onto the device ensuring you line it up against the top edge as shown below. Once aligned, lower the

Enclosure into place on to the Device. Finally, adjust the position of the Enclosure so that the gaps are even on all sides, and none of the sides are catching when you press the Enclosure down flat.



3. **Fasten the Enclosure** - Using a 5IP (Torx-Plus) driver, install the 4 screws into the foot wells on the Enclosure. Each screw should be tightened until snug, and then turned another 45-degrees (1/8th turn) to fully fasten. Adjust the position of the Enclosure as you go to ensure even gaps around the perimeter.

Caution: Ensure that the Enclosure is not stuck on a ledge as you are installing the screws. Shift the Enclosure as needed to avoid this condition.

4. **Install the Feet** – Refer to the [Procedure – Installation \(Feet\)](#) section of this document for detailed instructions.

Removable Solid-State Drive Replacement

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

Required Tools

- Plastic Opening Pick
- Soft ESD-Safe Mat
- 5IP (Torx-Plus) Driver
- Anti-Static wrist strap (1M Ohm resistance)
- USB drive loaded with the Surface Diagnostic Toolkit

Primary Components

- Removable Solid-State Drive (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)
 - M1246215 Screws x 1 (Solid-State Drive) **5IP**
 - M1234550 Tape x 1 (SSD)
 - M1318579 Spacer x 1 (SSD)

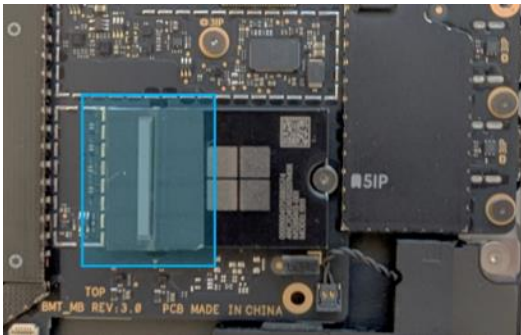
Additional Components (Ordered Separately)

- Feet (Refer to the Illustrated Service Parts List)

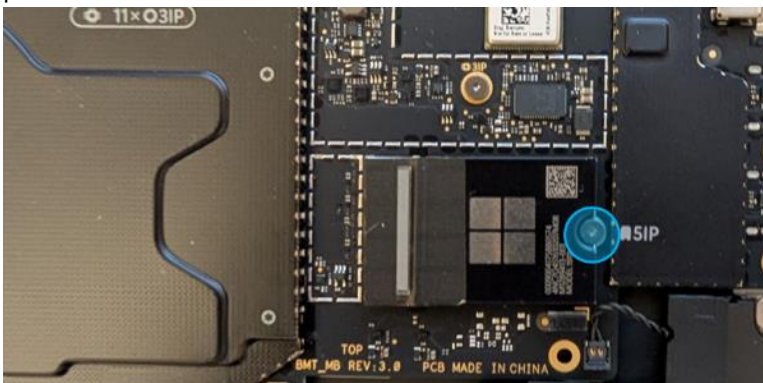
Procedure – Removal (Removable Solid-State Drive)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.

3. **Remove the Enclosure** – Refer to the [Procedure – Removal \(Enclosure\)](#) section of this document for detailed instructions.
4. **Remove the tape** - Carefully peel up and completely remove the tape covering the Removable Solid-State Drive and the corresponding receptacle. The gray spacer will also be removed along with the tape. Clean the top surfaces of the Removable Solid-State Drive and receptacle with IPA to remove any residual adhesive.



5. **Remove the Removable Solid-State Drive** – Using a 5IP (Torx-Plus) driver, remove the screw (5IP) holding the Removable Solid-State Drive onto the Motherboard. Lift the drive out of the device and place it on a soft ESD-Safe mat.



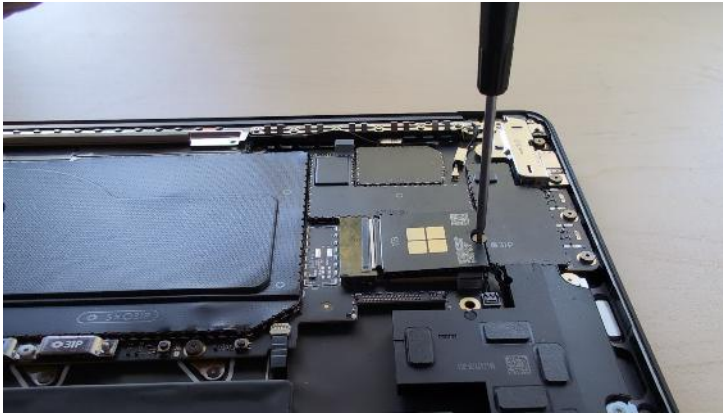
Procedure – Installation (Removable Solid-State Drive)

1. **Insert the Removable Solid-State Drive** – Carefully insert the connector end of the Removable Solid-State Drive into the receptacle on the motherboard while keeping the Removable Solid-State Drive as close as possible to horizontal.

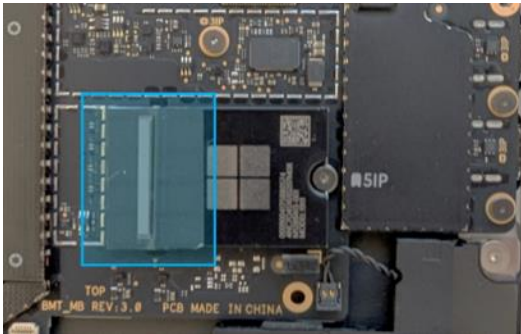
CAUTION: Ensure that the Speaker wire does not get caught or trapped when fastening the Removable Solid-State Drive to the Motherboard.



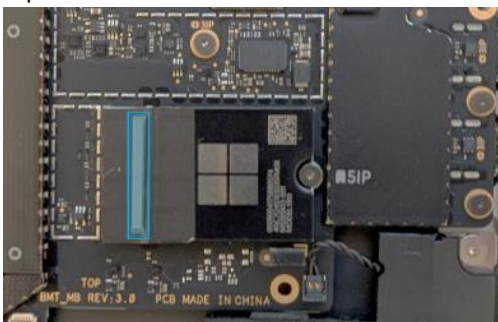
2. **Install Removable Solid-State Drive screw** – Using a 5IP (Torx-Plus) driver, install the 1 new rSSD screw (5IP) until the screw is just snug. Then turn the screw an additional 45-degrees (1/8th turn) until screw is fully fastened.



3. **Install SSD Tape on Motherboard Receptacle** – Carefully place a new tape on the receptacle, making sure the cutout on the tape matches the shape of the raised flange on the receptacle.



4. **Install Spacer** – Place a new spacer on top of the Black Tape, parallel and as close as possible to the edge of the connector receptable. Ensure that the spacer is centered lengthwise within the black tape.



5. **Install Enclosure** - Refer to the [Procedure – Installation \(Enclosure\)](#) section of this document for detailed instructions.
6. **Power on device** – Carefully place the device with the screen side facing up. Connect the device to a power supply and open the Display.
7. **Image the device** – Reinstall the operating system and all drivers/firmware by using a USB-drive containing the latest Surface BMR for your model. Please see [the Software Tools – Diagnostic, Calibration, and Troubleshooting](#) section for links to instructions on how to get the latest image and install it.

8. **Run the Surface Diagnostic Toolkit (SDT)** – With Windows installed and sitting at the desktop, insert the USB drive containing SDT and launch the program. Run all diagnostics to ensure the device is functioning as expected before moving forward.
9. **Power off the device** – Once the SDT tests have completed, power down the device and close the display. Invert the device so that the bottom of the device is facing up.
10. **Install the Feet** - Refer to the [Procedure – Installation \(Feet\)](#) section of this document for detailed instructions.

Battery Replacement

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

WARNING: Prior to handling the battery, the operator must remove all jewelry, wear gloves and safety glasses, and have a bucket of sand prepared in case of any battery event.

WARNING: In the case of a battery event, submerge the entire device in sand. **DO NOT** attempt to pick up the device.

Required Tools

- Plastic Opening Pick
- Nylon Spudger
- ESD-Safe Tweezers
- Soft ESD-Safe Mat
- 5IP (Torx-Plus) Driver
- 3IP (Torx-Plus) Driver
- Anti-Static wrist strap (1M Ohm resistance)
- USB drive loaded with the Surface Diagnostic Toolkit
- 4 Gallon Bucket
- 2.0 Gallons of Clean Sand

Primary Components

- Battery (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)
 - M1246215 Screws x 1 (Solid-State Drive) **5IP**
 - M1266593 Screws x 2 (Battery FPC Bracket) **3IP - 1**
 - M1243031 (13" Laptops) or M1272782 (15" Laptops) Screws x 8 (Battery) **5IP - 2**
 - M1234550 Tape x 1 (SSD)
 - M1318579 Spacer x 1 (SSD)

Additional Components (Ordered Separately)

- Feet (Refer to the Illustrated Service Parts List)

Procedure – Preparation (Battery)

Important: This section is only for instances where you are replacing the battery. If the battery is being re-used, then this section is not required.

1. **Connect USB** – Connect USB with the Surface Diagnostic Toolkit (SDT) loaded to an available USB port on the device under repair.
2. **Power on device** – Connect a power supply to the device. Press the power button on the device to power the device on. Allow it to boot to the Windows Desktop before continuing.
3. **Launch SDT** – From the Windows Desktop, use Windows Explorer to navigate to the USB drive. Select the SDT executable (.exe) to launch the Surface Diagnostic Toolkit.
4. **Run Battery Repair (Setup)** – From the SDT launch screen, select **Repair** from the drop-down menu. Next, select **Repair Setup and Validation** to enter the selection screen. Run the **Battery Repair (Setup)** to put your device into repair mode. Follow all on-screen instructions and allow the device to shut down when prompted. Disconnect the Power Supply and remove the USB drive before proceeding forward.

Procedure – Removal (Battery)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.
3. **Remove the Enclosure** – Refer to the [Procedure – Removal \(Enclosure\)](#) section of this document for detailed instructions.
4. **Remove the Removable Solid-State Drive** - Refer to the [Procedure – Removal \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
5. **Remove the Battery Connector Metal Bracket** – Using a 3IP (Torx-Plus) driver, remove the 2 screws ( 3IP) securing the metal bracket to the motherboard. Lift the metal bracket out of the device to expose the Battery FPC.



6. **Disconnect the Battery FPC** - Using a Nylon Spudger, pry the Battery FPC connector, starting from the side of the connector, from the Motherboard.



7. **Remove the Battery Screws** - Using a 5IP (Torx-Plus) driver, remove the 8 screws securing the Battery.



8. **Remove the battery from the device** - Using your hands, grab the battery at the four points identified below and carefully lift the battery out of the device. Place the battery on a clean ESD-Safe mat free of any debris.



WARNING: Only handle the battery by the plastic frame. Bending, twisting, or impacting the battery may damage the battery, the device, and/or result in severe personal injury or property damage. Always use two hands when handling the battery.

Important: Place the battery somewhere where the battery cannot accidentally be contacted or damaged. **DO NOT** place anything on top of the battery.

Important: When disposing of the battery, ensure you are recycling according to local laws.

Important: The Motherboard Module and Battery are extremely sensitive to ESD and can be easily damaged. It is critical that you ensure proper grounding before performing any work on these parts.

WARNING: In the instance of a battery event, submerge the entire device in a 4-gallon bucket filled with 2.0 gallons of clean sand. Ensure the entire device is submerged. **DO NOT** attempt to pick up the device.

Procedure – Installation (Battery)

1. **Pre-installation device inspection** – Check the device interior for any loose articles that may be present.
 - a. Check and remove any foreign objects that the magnets may have attracted.
 - b. Pay special attention to the magnetized areas around the edges of the interior.
 - c. Verify that all removed screws are accounted for and have not been misplaced inside the device.
 - d. Loose screws should never be stored on the magnetic areas of the bucket.

Important: Verify the battery's condition. Batteries exhibiting any damage indicated in the Lithium-Ion Battery Inspection section must be replaced.

2. **Insert the Battery** - Using the attached loops, carefully lower the battery into the new device.

Important: Only handle new batteries with the plastic loops that come attached. If reusing a battery, handle by the frame as indicated in the battery removal instructions. Bending, twisting, or impacting the battery may damage the battery, the device, and/or result in severe personal injury or property damage. Always use two hands when handling the battery.

3. **Install the Battery screws** - Using a 5IP screwdriver, install the 8 battery frame screws (🔩5IP - 2) until the screws are just snug, then tighten each by an additional 45-degrees (1/8th turn).

Important: Do not overtighten the screws on the battery frame or battery. If the frame is cracked, the battery must not be used.

4. **Assemble the Battery FPC and FPC Bracket** - Assemble the Battery FPC to the Motherboard. Using a 3IP (Torx-Plus) driver, install 2 FPC Bracket screws (🔩3IP - 1) until the screws are just snug. Turn each screw an additional 45-degrees (1/8th turn) until fully fastened.
5. **Install the Removable Solid-State Drive** – Refer to [Procedure – Installation \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
6. **Install the Enclosure** – Refer to [Procedure – Installation \(Enclosure\)](#) section of this document for detailed instructions.

Procedure – Finalize (Battery)

1. **Power on Device** – Connect a Power Supply to the device and power it on until it reaches the Windows Desktop.
2. **Connect USB** – Connect USB with the Surface Diagnostic Toolkit (SDT) loaded to an available USB port on the device under repair.
3. **Launch SDT** – From the Windows Desktop, use Windows Explorer to navigate to the USB drive. Select the SDT executable (.exe) to launch the Surface Diagnostic Toolkit.
4. **Allow the Battery to charge** – With the device connected to a power supply, allow the battery to charge until the battery icon in Windows reads at least 50% remaining battery charge.
5. **Run Battery Authentication** – From the SDT launch screen, select **Repair** from the drop-down menu. Next, select **Repair Setup and Validation** to enter the selection screen. Select the **Battery Repair (Validation)** tool and follow the on-screen prompts until a successful authentication is completed.

Important: Battery authentication requires a stable internet connection and the latest version of the [Surface Management Extension](#). If the battery validation tool fails or is not detected properly, install the Surface Management Extension, reboot the device, and try again with a new internet connection. If failures continue, reach out to Microsoft Support.

6. **Run the Surface Diagnostic Toolkit (SDT)** – Run all diagnostics to ensure the device is functioning as expected before moving forward.
7. **Install Feet** – Refer to [Procedure – Installation \(Feet\)](#) section of this document for detailed instructions.

Audio Jack Replacement

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

Required Tools

- Plastic Opening Pick
- Nylon Spudger
- ESD-Safe Tweezers
- Soft ESD-Safe Mat
- 6IP (Torx-Plus) Driver
- 5IP (Torx-Plus) Driver
- 3IP (Torx-Plus) Driver
- Anti-Static wrist strap (1M Ohm resistance)
- USB drive loaded with the Surface Diagnostic Toolkit

Primary Components

- Audio Jack (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)
 - M1246215 Screws x 1 (Solid-State Drive) **5IP**
 - M1265600 Screws x 1 (Hinge & Chassis) **6IP**
 - M1212080 Screws x 1 (Upper Audio Jack Bridge) **3IP - 1**
 - M1330715 Screws x 1 (Lower Audio Jack Bridge)
 - M1263960 Screws x 1 (Audio Jack) **3IP - 2**
 - M1234550 Tape x 1 (SSD)
 - M1318579 Spacer x 1 (SSD)

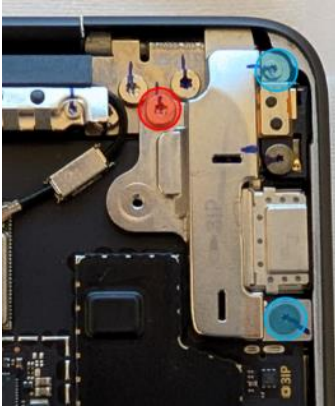
Additional Components (Ordered Separately)

- Feet (Refer to the Illustrated Service Parts List)

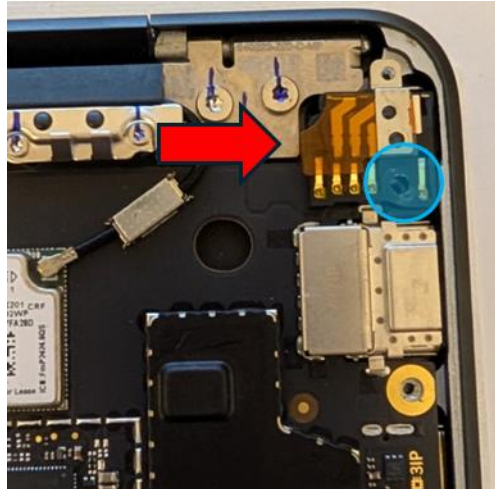
Procedure – Removal (Audio Jack)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.
3. **Remove the Enclosure** – Refer to the [Procedure – Removal \(Enclosure\)](#) section of this document for detailed instructions.
4. **Remove the Removable Solid-State Drive** - Refer to the [Procedure – Removal \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.

5. **Remove the Audio Jack Bridge** – Using a 3IP (Torx-Plus) and 6IP (Torx-Plus) driver, remove the 3 screws securing the Audio Jack Bridge to the interior. The 6IP screw is circled in red. Lift the Audio Jack Bridge out of the device.



6. **Remove the Audio Jack** –
 - a. Using a 3IP (Torx-Plus) screwdriver, remove the 1 screw securing the Audio Jack to the Motherboard.
 - b. Lift the Audio Jack out of the device by pulling on the orange tab.



Procedure – Installation (Audio Jack)

1. **Install the Audio Jack** –
 - a. Place the Audio Jack into its position on the Motherboard.
 - b. Using a 3IP (Torx-Plus) driver to install a new Audio Jack screw (3IP - 2) until just snug. Then turn the screw an additional 45-degrees (1/8th turn) until fully fastened.
2. **Install the Audio Jack Bridge** - Install the previously removed Audio Jack Bridge into its position over the Audio Jack. Using a 3IP (Torx-Plus) and 6IP (Torx-Plus) driver, install 3 new screws (1 x 3IP - 1, 1 x 6IP, 1 x M1330715). All screws should be tightened until just snug, and then turned another 45-degrees (1/8th turn) until fully fastened.
3. **Install the Removable Solid-State Drive** – Refer to [Procedure – Installation \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
4. **Install the Enclosure** – Refer to [Procedure – Installation \(Enclosure\)](#) section of this document for detailed instructions.
5. **Power on device** – Carefully place the device with the screen side facing up. Connect the device to a power supply and open the Display.

6. **Run the Surface Diagnostic Toolkit (SDT)** – Sitting at the desktop, insert the USB drive containing SDT and launch the program. Run all diagnostics to ensure the device is functioning as expected before moving forward.
7. **Power off the device** – Once the SDT tests have completed, power down the device and close the display. Invert the device so that the bottom of the device is facing up.
8. **Install the Feet** - Refer to the [Procedure – Installation \(Feet\)](#) section of this document for detailed instructions.

Right Speaker Replacement

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

Note: The Right Speaker is on the Left when the device is oriented for repair.

Required Tools

- Plastic Opening Pick
- Nylon Spudger
- ESD-Safe Tweezers
- Soft ESD-Safe Mat
- 5IP (Torx-Plus) Driver
- 3IP (Torx-Plus) Driver
- Anti-Static wrist strap (1M Ohm resistance)
- USB drive loaded with the Surface Diagnostic Toolkit

Primary Components

- Right Speaker (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)
 - M1246215 Screws x 1 (Solid-State Drive) **5IP**
 - M1211914 Screws x 2 (Speaker) **3IP**
 - M1167842 Tape x 1 (Right Speaker Tape)
 - M1234550 Tape x 1 (SSD)
 - M1318579 Spacer x 1 (SSD)

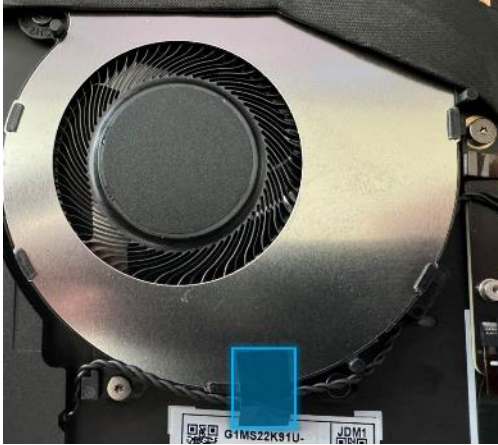
Additional Components (Ordered Separately)

- Feet (Refer to the Illustrated Service Parts List)

Procedure – Removal (Right Speaker)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.
3. **Remove the Enclosure** – Refer to the [Procedure – Removal \(Enclosure\)](#) section of this document for detailed instructions.

4. **Remove the Removable Solid-State Drive** - Refer to the [Procedure – Removal \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
5. **Remove the Speaker tape** – Remove the tape and clean the surface with IPA to remove any residual adhesive. **This is only required for 13” Laptops.**



6. **Remove the Right Speaker screws** – Using a 3IP (Torx-Plus) driver to remove the 2 screws securing the Right Speaker (which is on the left side when working on the device) to the Chassis.



7. **Remove the Right Speaker** – Remove the Right Speaker from the Chassis and de-route the cable. To remove the connector from the Motherboard, pull up vertically on the wire until the connector comes free.

Procedure – Installation (Right Speaker)

1. **Install the Right Speaker** - Place the new Right Speaker into the Chassis. Using a 3IP (Torx-Plus) driver, install 2 new screws (3IP). All screws should be installed until just snug, and then turned another 45-degrees (1/8th turn) until fully fastened.

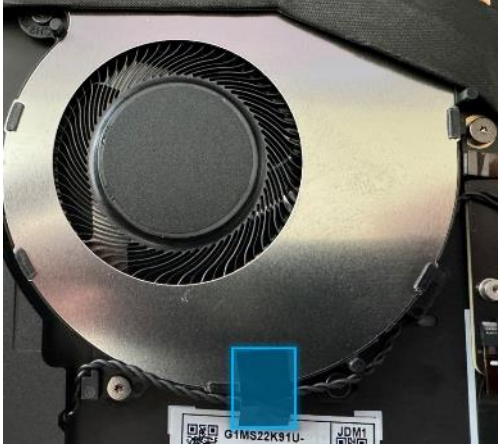


2. **Route and connect the Right Speaker wire** – Route the Right Speaker wire as shown below. Install the speaker connector into the receptacle on the Motherboard by pressing vertically until a snap is felt.

NOTE: The connector will have 2 visible gold contacts if it is oriented in the correct direction.



3. **Place new Speaker Tape** - Place a new Tape over the speaker wire as shown below. **This is only required on 13" Laptops.**



4. **Install the Removable Solid-State Drive** – Refer to [Procedure – Installation \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
5. **Install the Enclosure** – Refer to [Procedure – Installation \(Enclosure\)](#) section of this document for detailed instructions.
6. **Power on device** – Carefully place the device with the screen side facing up. Connect the device to a power supply and open the Display.
7. **Run the Surface Diagnostic Toolkit (SDT)** – Sitting at the desktop, insert the USB drive containing SDT and launch the program. Run all diagnostics to ensure the device is functioning as expected before moving forward.
8. **Power off the device** – Once the SDT tests have completed, power down the device and close the display. Invert the device so that the bottom of the device is facing up.
9. **Install the Feet** - Refer to the [Procedure – Installation \(Feet\)](#) section of this document for detailed instructions.

Left Speaker Replacement

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

Required Tools

- Plastic Opening Pick
- Nylon Spudger
- ESD-Safe Tweezers
- Soft ESD-Safe Mat
- 5IP (Torx-Plus) Driver
- 3IP (Torx-Plus) Driver
- Anti-Static wrist strap (1M Ohm resistance)
- USB drive loaded with the Surface Diagnostic Toolkit

Primary Components

- Right Speaker (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)
 - M1246215 Screws x 1 (Solid-State Drive) **5IP**
 - M1211914 Screws x 2 (Speaker) **3IP**

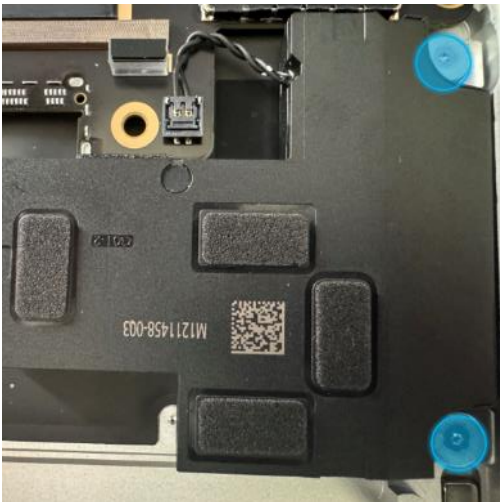
- M1234550 Tape x 1 (SSD)
- M1318579 Spacer x 1 (SSD)

Additional Components (Ordered Separately)

- Feet (Refer to the Illustrated Service Parts List)

Procedure – Removal (Left Speaker)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.
3. **Remove the Enclosure** – Refer to the [Procedure – Removal \(Enclosure\)](#) section of this document for detailed instructions.
4. **Remove the Removable Solid-State Drive** - Refer to the [Procedure – Removal \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
5. **Remove the Left Speaker Screws** – Using a 3IP (Torx-Plus), remove the 2 screws securing the Left Speaker, located on the Right side of the device when working on it, to the Chassis.

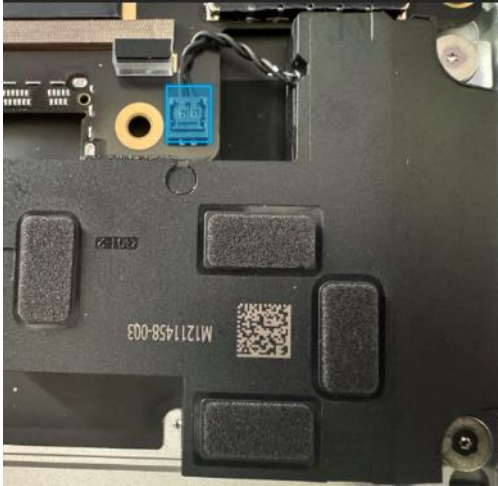


6. **Remove the Left Speaker** – Remove the Left Speaker from the Chassis. To Remove the connector from the Motherboard, pull up vertically on the wires until the connector comes free.

Procedure – Installation (Left Speaker)

1. **Install the Left Speaker** – Place the new Left Speaker into the Chassis. Using a 3IP (Torx-Plus) driver to install 2 new screws (3IP). All screws should be installed until just snug, and then turned another 45-degrees (1/8th turn) until fully fastened.
2. **Connect the Left Speaker** – Insert the speaker connector into the receptable on the Motherboard by pressing vertically until a snap is felt.

NOTE: The connector will have 2 visible gold contacts if it is oriented in the correct direction.



3. **Install the Removable Solid-State Drive** – Refer to [Procedure – Installation \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
4. **Install the Enclosure** – Refer to [Procedure – Installation \(Enclosure\)](#) section of this document for detailed instructions.
5. **Power on device** – Carefully place the device with the screen side facing up. Connect the device to a power supply and open the Display.
6. **Run the Surface Diagnostic Toolkit (SDT)** – Sitting at the desktop, insert the USB drive containing SDT and launch the program. Run all diagnostics to ensure the device is functioning as expected before moving forward.
7. **Power off the device** – Once the SDT tests have completed, power down the device and close the display. Invert the device so that the bottom of the device is facing up.
8. **Install the Feet** - Refer to the [Procedure – Installation \(Feet\)](#) section of this document for detailed instructions.

Micro SD Reader Replacement

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

Note: Only 15” Laptops will have a Micro SD Reader.

Required Tools

- Plastic Opening Pick
- Nylon Spudger
- ESD-Safe Tweezers
- Soft ESD-Safe Mat
- 5IP (Torx-Plus) Driver
- 3IP (Torx-Plus) Driver
- Anti-Static wrist strap (1M Ohm resistance)
- USB drive loaded with the Surface Diagnostic Toolkit

Primary Components

- Micro SD (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)

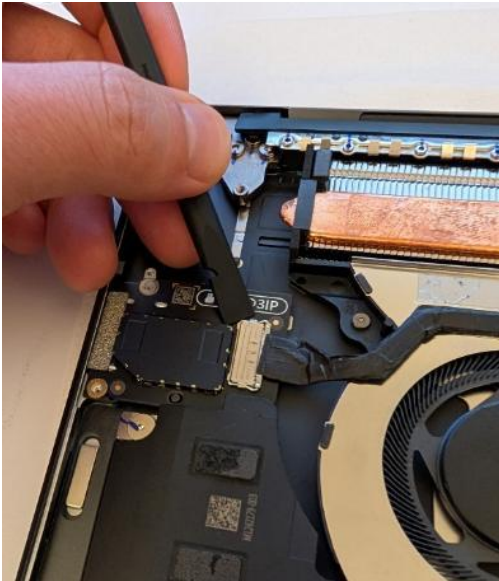
- M1246215 Screws x 1 (Solid-State Drive) **5IP**
- M1235998 Screws x 2 (Micro SD) **3IP**
- M1234550 Tape x 1 (SSD)
- M1318579 Spacer x 1 (SSD)

Additional Components (Ordered Separately)

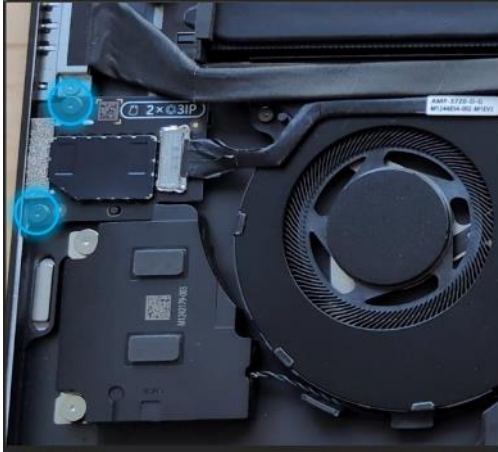
- Feet (Refer to the Illustrated Service Parts List)

Procedure – Removal (Micro SD Reader)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.
3. **Remove the Enclosure** – Refer to the [Procedure – Removal \(Enclosure\)](#) section of this document for detailed instructions.
4. **Remove the Removable Solid-State Drive** - Refer to the [Procedure – Removal \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
5. **Disconnect the Micro SD Reader Cable** – Using a Nylon Spudger, flip up the latch on the Micro SD Reader board and disconnect the connector.



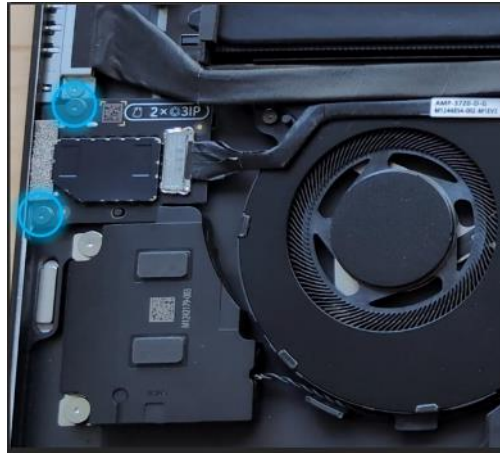
6. **Remove the Micro SD Reader** – Using a 3IP (Torx-Plus) driver, remove the 2 screws holding the Micro SD Reader into the Chassis and lift the Micro SD Reader out of the device.



Procedure – Installation (Micro SD Reader)

1. Install the Micro SD Reader –

- a. Carefully place the new Micro SD Reader into the device chassis.
- b. Verify that the reader is aligned with the opening on the outside of the device chassis.
- c. Using a 3IP (Torx-Plus) driver, install 2 new screws (**3IP**) until just snug and then turn each another 45-degrees (1/8th turn) until fully fastened.



2. **Connect the Micro SD Reader Cable** - Insert the Micro SD Reader cable into the receptacle on the Micro SD Reader board and close the latch. There should be a click if the connector is inserted correctly, and the latch is fully closed.
3. **Install the Removable Solid-State Drive** – Refer to [Procedure – Installation \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
4. **Install the Enclosure** – Refer to [Procedure – Installation \(Enclosure\)](#) section of this document for detailed instructions.
5. **Power on device** – Carefully place the device with the screen side facing up. Connect the device to a power supply and open the Display.
6. **Run the Surface Diagnostic Toolkit (SDT)** – Sitting at the desktop, insert the USB drive containing SDT and launch the program. Run all diagnostics to ensure the device is functioning as expected before moving forward.
7. **Power off the device** – Once the SDT tests have completed, power down the device and close the display. Invert the device so that the bottom of the device is facing up.

8. **Install the Feet** - Refer to the [Procedure – Installation \(Feet\)](#) section of this document for detailed instructions.

Display Assembly Replacement

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

Required Tools

- Plastic Opening Pick
- Nylon Spudger
- ESD-Safe Tweezers
- Soft ESD-Safe Mat
- 6IP (Torx-Plus) Driver
- 5IP (Torx-Plus) Driver
- 3IP (Torx-Plus) Driver
- Anti-Static wrist strap (1M Ohm resistance)
- USB drive loaded with the Surface Diagnostic Toolkit

Primary Components

- Display Assembly (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)
 - M1246215 Screws x 1 (Solid-State Drive) **5IP**
 - M1265600 Screws x 8 (Hinge & Chassis) **6IP**
 - M1274578 Screws x 12 (13" Laptops) or x 14 (15" Laptops) (Antenna) **3IP**
 - M1212080 Screws x 1 (Upper Audio Jack Bridge) **3IP - 1**
 - M1330715 Screws x 1 (Lower Audio Jack Bridge)
 - M1263960 Screws x 1 (Audio Jack) **3IP - 2**
 - M1235995 Screws x 1 (Motherboard) **3IP - 1**
 - M1265416 Shield x 2 (Coax Cable Lid)
 - M1271924 Shield x 1 (T3 Shield)
 - M1288973 Foam x 1 (T3 Shield Foam #1)
 - M1288974 Foam x 1 (T3 Shield Foam #2)
 - M1328581 Tape x 1 (Display Assembly FPC Tape)
 - M1328578 Tape x 1 (Display Assembly FPC Tape)
 - M1291196 Tape x 1 (Display Assembly FPC Tape)
 - M1234550 Tape x 1 (SSD)
 - M1318579 Spacer x 1 (SSD)

Additional Components (Ordered Separately)

- Feet (Refer to the Illustrated Service Parts List)

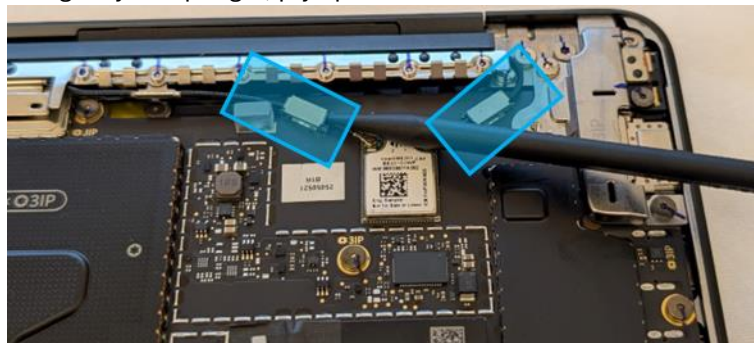
Procedure – Preparation (Display Assembly)

Important: This section is only for instances where you are replacing the Display. If the Display is being re-used, then this section is not required. If Display is unusable due to damage or fault, connect an external monitor to the device to perform these steps.

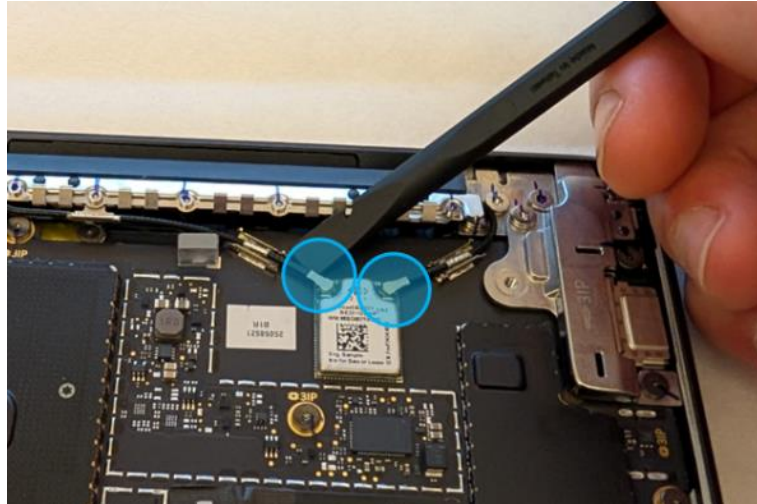
1. **Connect USB** – Connect USB with the Surface Diagnostic Toolkit (SDT) loaded to an available USB port on the device under repair.
2. **Power on device** – Connect a power supply to the device. Press the power button on the device to power the device on. Allow it to boot to the Windows Desktop before continuing.
3. **Launch SDT** – From the Windows Desktop, use Windows Explorer to navigate to the USB drive. Select the SDT executable (.exe) to launch the Surface Diagnostic Toolkit.
4. **Run Touch Display Setup** – From the SDT launch screen, select **Repair** from the drop-down menu. Next, select **Repair Setup and Validation** to enter the selection screen. Run the **Touch Display (Setup)** tool to prepare your device for Display replacement. Follow all on-screen instructions and allow the device to shut down when prompted. Disconnect the Power Supply and remove the USB drive before proceeding forward.

Procedure – Removal (Display Assembly)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.
3. **Remove the Enclosure** – Refer to the [Procedure – Removal \(Enclosure\)](#) section of this document for detailed instructions.
4. **Remove the Removable Solid-State Drive** - Refer to the [Procedure – Removal \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
5. **Remove the Audio Jack** - Refer to the [Procedure – Removal \(Audio Jack\)](#) section of this document for detailed instructions.
6. **Remove the Antenna** –
 - a) Using a Nylon Spudger, pry up the 2 Coax Cable Lids.



- b) Using a Nylon Spudger, disconnect the 2 Coax Cables.



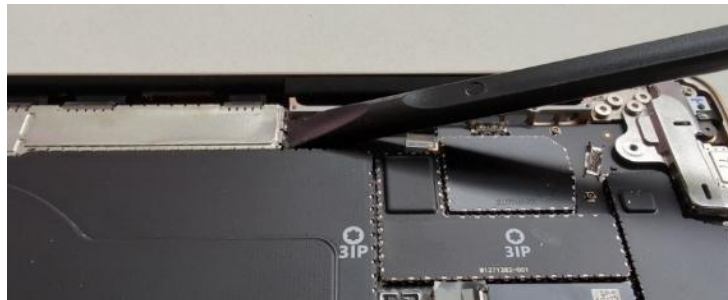
- c) Using a 3IP (Torx-Plus) driver, remove the 12 or 14 screws from the antenna.



- d) Lift the Antenna out of the Chassis.

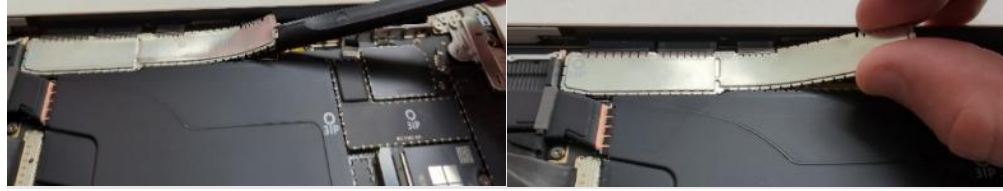
7. Remove the T3 Shield Lid –

- a) Using a Nylon Spudger, pry up the T3 Shield Lid starting with the right edge.



NOTE: Removing the Motherboard screw next to the T3 Shield can give better access for removal of the shield.

- b) Slide the Nylon Spudger under the shield moving right to left. Pause and shift the shield up and down as you work your way left as the shield will get stuck on its latches.



Caution: Ensure that the shield does not damage the Display FPC's during removal.

8. **Disconnect the Display Assembly FPCs** - Using a Nylon Spudger, pry up the connectors from the side and gently wiggle them free. If the connectors start giving resistance as they are being pried up, lightly push the edge being pried back down.

Caution: Do not force the connector if it starts to give resistance. Rock the connector back in the other direction to remove.



9. **Reorient the Device –**
 - a) Open the Display Assembly to 90 Degrees.
 - b) Place the backside of the Display Assembly on the ESD-Safe Mat with the screen and keyboard facing up.



10. **Remove the Left Hinge Screws –** With one hand holding the device still, use a 6IP (Torx-Plus) driver to remove the 3 screws on the Left Hinge.



11. **Remove the Right Hinge Screws** - With one hand holding the device still, use a 6IP (Torx-Plus) driver to remove the 4 screws on the Right Hinge.
12. **Remove the Display Assembly from the Device** - Carefully lift the Chassis from the Display Assembly. Place the Display Assembly on an ESD-Safe mat.

Procedure – Installation (Display Assembly)

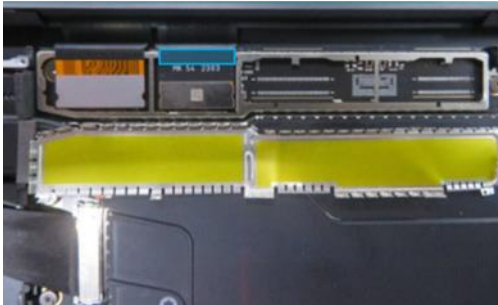
1. **Orient Display Assembly** – Place the backside of the new Display Assembly on the ESD mat with each of the hinges set at 90-degree angles.
2. **Install the new Display Assembly** – Carefully align the hinges into the packets on the device Enclosure.

Important: Ensure the Enclosure does not impact on the display assembly glass.

3. **Pre-fasten the right hinge screws** - Using a 6IP (Torx-Plus) driver, install 4 new right side hinge screws ( 6IP) and tighten until it is just snug.
4. **Pre-fasten the left hinge screws** - Using a 6IP (Torx-Plus) driver, install 3 new left side hinge screws ( 6IP) and tighten until just snug.
5. **Check alignment –**
 - a. Close the device.
 - b. Loosen all 7 hinge screws 90 degrees (1/4th turn).
 - c. Adjust the alignment until the gap between the Display Assembly and the Chassis is as even as possible.



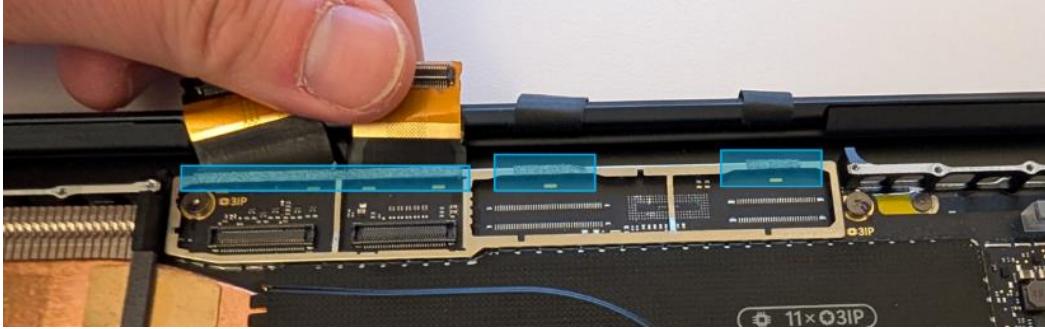
- d. Tighten down 1 screw on the left hinge until snug. Adjust to the right side so that the Chassis surface is flush with the Display Assembly.
- e. Tighten down 1 screw on the right hinge. Loosen the screw that was tightened down on the left hinge. Adjust the left side so that the Chassis surface is flush with the Display Assembly.
- f. Repeat as necessary until the left and right gaps are even and the back surfaces are flush.
6. **Tighten all hinge screws** - Tighten all 7 hinge screws until they are snug, and then tighten an additional 90 degrees (1/4 turn) to ensure they are securely fastened.
7. **Place the Display Assembly FPC Tape (as needed)** – If re-using a Display Assembly, place the Display Assembly FPC Tape (M1291196) to the Display Assembly FPC as shown in blue.



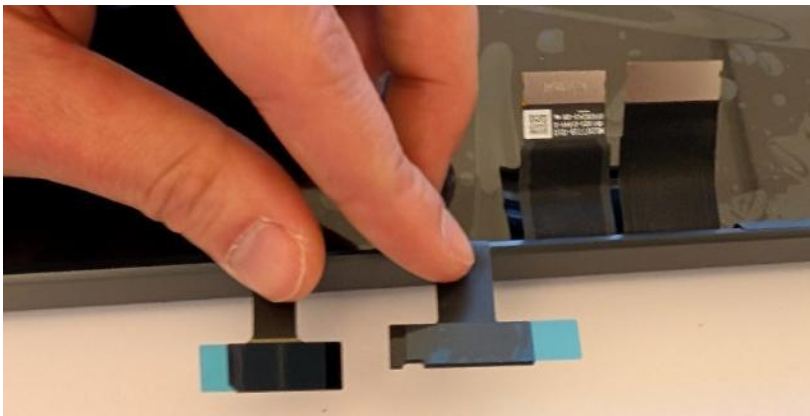
If re-using a Display Assembly and 2 of the FPCs on the Display Assembly do not have connectors, remove the existing tapes and clean the FPC with IPA to remove any adhesive residue. Then install the new Display Assembly FPC Tapes (M1328578 and M1328581).



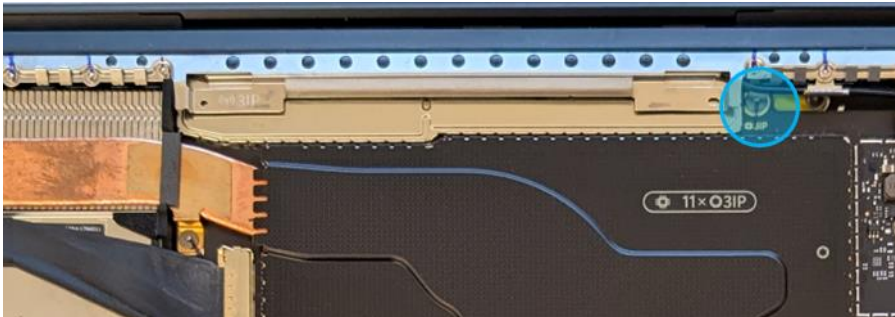
8. **Install the T3 Shield Foams (as needed)** – If the foams in the indicated areas were damaged during the removal of the T3 shield, completely remove them and clean the T3 shield with IPA to remove any adhesive residue. Then install the new foams.



9. **Assemble the Display Assembly FPCs to the Motherboard** - Assemble the Display FPCs to the Motherboard. If installing a new Display Assembly, remove the protective blue liners from the FPC and adhere the FPCs to the connectors on the Motherboard. Ensure that the FPCs are bent under the Motherboard. **Do not crease the FPCs**



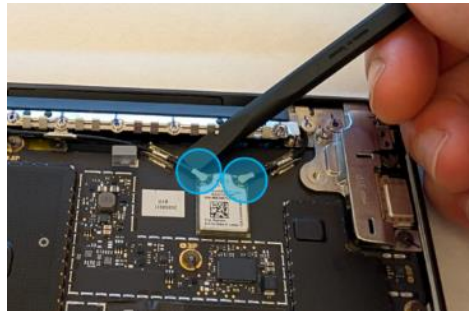
10. **Install the Motherboard screw** – If this screw was removed during the disassembly process, using a 3IP (Torx-Plus) driver, install 1 new Motherboard screw (3IP) until just snug and seated, and then turn another 45-degrees (1/8th turn) until fully fastened.



11. **Install the Antenna** –
 - a. Install the previously removed Antenna. Using a 3IP (Torx-Plus) driver, install 12 or 14 new screws (3IP). All screws should be installed until just snug, then turned another 45-degrees (1/8th turn) until fully fastened.



- b. Connect the 2 Coax Cables to the Motherboard by aligning each with the socket, and pressing down until a click is felt.



- c. Install 2 new Coax Cable Lids to the Motherboard over the Coax Cables. Align and press into place until a click is felt.



12. **Install the Audio Jack** – Refer to [Procedure – Installation \(Audio Jack\)](#) section of this document for detailed instructions.
13. **Install the Removable Solid-State Drive** – Refer to [Procedure – Installation \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
14. **Install the Enclosure** – Refer to [Procedure – Installation \(Enclosure\)](#) section of this document for detailed instructions.
15. **Power on device** – Carefully place the device with the screen side facing up. Connect the device to a power supply and open the Display.

Procedure – Finalize (Display Assembly)

Important: This section is only for instances where you are replacing the Display. If the Display is being re-used, then this section is not required. If Display is unusable due to damage or fault, connect an external monitor to the device to perform these steps.

1. **Connect USB** – Connect USB with the Surface Diagnostic Toolkit (SDT) loaded to an available USB port on the device under repair.
2. **Launch SDT** – From the Windows Desktop, use Windows Explorer to navigate to the USB drive. Select the SDT executable (.exe) to launch the Surface Diagnostic Toolkit.

3. **Run Touch Display Calibration** – From the SDT launch screen, select **Repair** from the drop-down menu. Next, select **Repair Setup and Validation** to enter the selection screen. Run the **Touch Display (Calibration)** tool to calibrate your new Display. Follow all on-screen instructions and allow the device to restart when prompted.

Important: If the calibration fails, reboot the device, and attempt again. If the failure continues, then the Display may be faulty and require replacement.

4. **Launch SDT** – Once the device has rebooted and is at the Windows Desktop, use Windows Explorer to navigate to the USB drive. Select the SDT executable (.exe) to launch the Surface Diagnostic Toolkit.
5. **Run the Surface Diagnostic Toolkit (SDT)** – Run all diagnostics to ensure the device is functioning as expected before moving forward.
6. **Install Feet** – Refer to [Procedure – Installation \(Feet\)](#) for steps to install Feet.

Surface Connect Replacement

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

Required Tools

- Plastic Opening Pick
- Nylon Spudger
- ESD-Safe Tweezers
- Soft ESD-Safe Mat
- 5IP (Torx-Plus) Driver
- 3IP (Torx-Plus) Driver
- Isopropyl alcohol (91% or greater)
- Cleaning swabs
- Anti-Static wrist strap (1M Ohm resistance)
- USB drive loaded with the Surface Diagnostic Toolkit

Primary Components

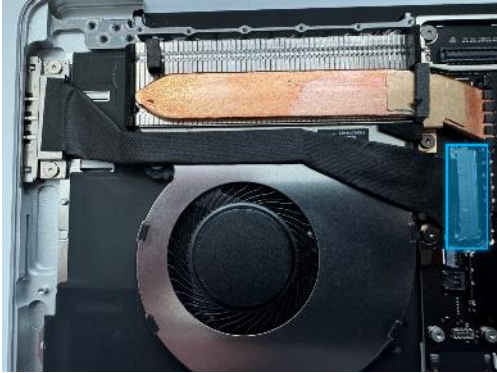
- Surface Connect (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)
 - M1246215 Screws x 1 (Solid-State Drive) **5IP**
 - M1277572 Screws x 2 (Surface Connect) **3IP**
 - M1301902 PSA x 1 (Surface Connect & Fan)
 - M1234550 Tape x 1 (SSD)
 - M1318579 Spacer x 1 (SSD)

Additional Components (Ordered Separately)

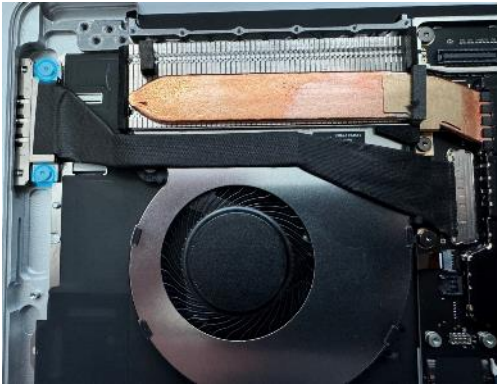
- Feet (Refer to the Illustrated Service Parts List)

Procedure – Removal (Surface Connect)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.
3. **Remove the Enclosure** – Refer to the [Procedure – Removal \(Enclosure\)](#) section of this document for detailed instructions.
4. **Remove the Removable Solid-State Drive** - Refer to the [Procedure – Removal \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
5. **Disconnect the Surface Connect connector** – Using a Nylon Spudger, flip up the latch on the Motherboard, and disconnect the Surface Connect connector from the Motherboard.



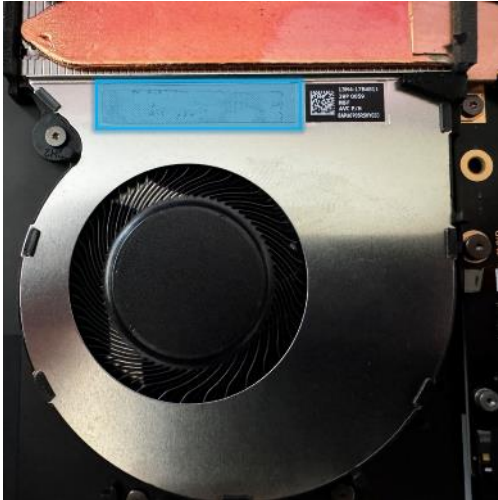
6. **Remove the Surface Connect Cable** - Using a 3IP (Torx-Plus) driver, remove the 2 screws securing the Surface Connect port to the Motherboard. Lift the cable out of the device and place it on an ESD-Safe mat.



7. **Remove residual adhesive** - Gently peel up the PSA (pressure-sensitive adhesive) that adheres the Surface Connect Cable to the Fan and remove. Clean up the surface with IPA and cleaning swabs until all residual adhesive has been removed and the surface is clean.

Procedure – Installation (Surface Connect)

1. **Install new PSA** - Place a new piece of PSA down onto the Fan as shown.



2. **Install the Surface Connect Cable –**

- a. Insert the connector of the Surface Connect Cable into the receptacle on the Motherboard and close the latch to secure it. There should be a click if the connector is inserted correct and the latch is fully closed.
- b. Toe in the Surface Connect connector into the device chassis ensuring that the plastic housing is flush with the outside surface of the chassis. The port should be fully aligned with the external opening.



- c. Using a 3IP (Torx-Plus) driver, install the 2 new Surface Connect screws (**3IP**) until the screw is just snug and seated, and then turn each another 45-degrees (1/8th turn) until fully fastened.
 - d. Lightly press the cable into the PSA to adhere the cable to the fan.
3. **Install the Removable Solid-State Drive** – Refer to [Procedure – Installation \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
 4. **Install the Enclosure** – Refer to [Procedure – Installation \(Enclosure\)](#) section of this document for detailed instructions.
 5. **Power on device** – Carefully place the device with the screen side facing up. Connect the device to a power supply and open the Display.
 6. **Run the Surface Diagnostic Toolkit (SDT)** – Sitting at the desktop, insert the USB drive containing SDT and launch the program. Run all diagnostics to ensure the device is functioning as expected before moving forward.
 7. **Power off the device** – Once the SDT tests have completed, power down the device and close the display. Invert the device so that the bottom of the device is facing up.
 8. **Install the Feet** - Refer to the [Procedure – Installation \(Feet\)](#) section of this document for detailed instructions.

Motherboard Replacement Process

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

Important: If replacing both the Motherboard and the Display Assembly – complete the Motherboard replacement prior to performing the Display Assembly Replacement to ensure proper part operation.

Required Tools

- Plastic Opening Pick
- Nylon Spudger
- ESD-Safe Tweezers
- Soft ESD-Safe Mat
- 6IP (Torx-Plus) Driver
- 5IP (Torx-Plus) Driver
- 3IP (Torx-Plus) Driver
- Isopropyl alcohol (91% or greater)
- Cleaning swabs
- Anti-Static wrist strap (1M Ohm resistance)
- USB drive loaded with the Surface Diagnostic Toolkit

Primary Components

- Motherboard (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)
 - M1266593 Screws x 2 (Battery FPC Bracket)  **3IP - 1**
 - M1243031 (13" Laptops) or M1272782 (15" Laptops) Screws x 8 (Battery)  **5IP - 2**
 - M1246215 Screws x 1 (Solid-State Drive)  **5IP**
 - M1265600 Screws x 1 (Hinge & Chassis)  **6IP**
 - M1274578 Screws x 12 (13" Laptops) or x 14 (15" Laptops) (Antenna)  **3IP**
 - M1212080 Screws x 1 (Upper Audio Jack Bridge)  **3IP - 1**
 - M1330715 Screws x 1 (Lower Audio Jack Bridge)
 - M1263960 Screws x 1 (Audio Jack)  **3IP - 2**
 - M1235995 Screws x 8 (Motherboard)  **3IP - 1**
 - M1263961 Screws x 2 (Motherboard)  **3IP - 2**
 - M1277573 Screws x 1 (Motherboard)  **3IP - 3**
 - M1265416 Shield x 2 (Coax Cable Lid)
 - M1330714 Shield x 1 (T1 Shield) (13" Laptops)
 - M1330703 Shield x 1 (T1 Shield) (15" Laptops)
 - M1271924 Shield x 1 (T3 Shield)
 - M1328581 Tape x 1 (Display Assembly FPC Tape)
 - M1328578 Tape x 1 (Display Assembly FPC Tape)
 - M1288973 Foam x 1 (T3 Shield Foam #1)
 - M1288974 Foam x 1 (T3 Shield Foam #2)
 - M1287120 Tape x 1 (Touchpad FPC Tape)

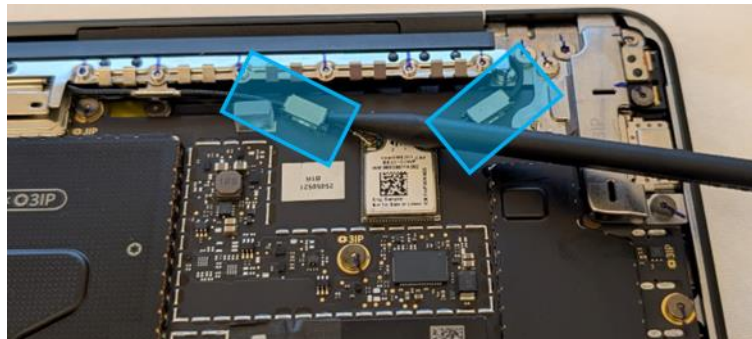
- M1019757 Syringe x 1 (Thermal Paste)
- M1301902 PSA x 1 (Surface Connect & Fan)
- M1304994 PSA x 1 (Micro SD Card Reader) (15" Laptops only)
- M1234550 Tape x 1 (SSD)
- M1318579 Spacer x 1 (SSD)

Additional Components (Ordered Separately)

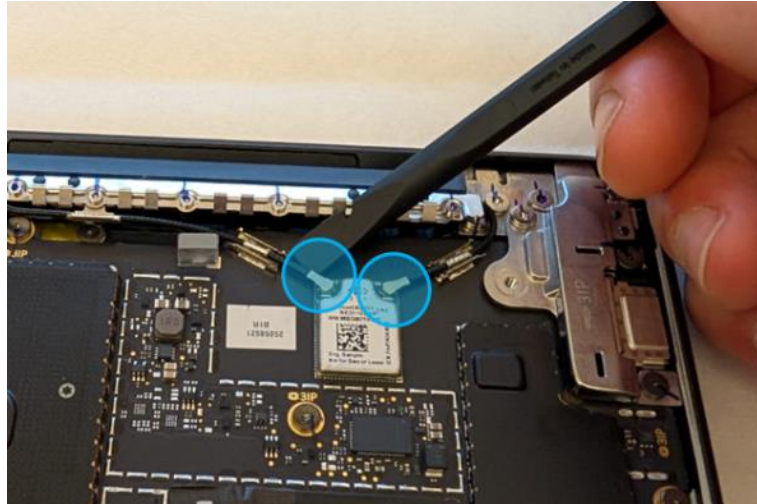
- Feet (Refer to the Illustrated Service Parts List)

Procedure – Removal (Motherboard)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.
3. **Remove the Enclosure** – Refer to the [Procedure – Removal \(Enclosure\)](#) section of this document for detailed instructions.
4. **Remove the Removable Solid-State Drive** - Refer to the [Procedure – Removal \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
5. **Remove the Audio Jack** - Refer to the [Procedure – Removal \(Audio Jack\)](#) section of this document for detailed instructions.
6. **Remove the Battery** – Refer to the [Procedure – Removal \(Battery\)](#) section of this document for detailed instructions.
7. **Remove the Antenna –**
 - a. Using a Nylon Spudger, pry up the 2 Coax Cable Lids.



- b. Using a Nylon Spudger, disconnect the 2 Coax Cables.



- c. Using a 3IP (Torx-Plus) driver, remove the 12 or 14 screws from the antenna.



- d. Lift the Antenna out of the Chassis.

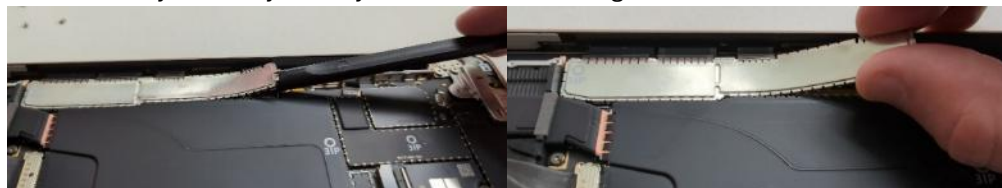
8. Remove the T3 Shield Lid –

- a. Using a Nylon Spudger, pry up the T3 Shield Lid starting with the right edge.



NOTE: Remove the Motherboard screw next to the T3 Shield can give better access for removal of the shield.

- b. Slide the Nylon Spudger under the shield moving right to left. Pause and shift the shield up and down as you work your way left as the shield will get stuck on its latches.



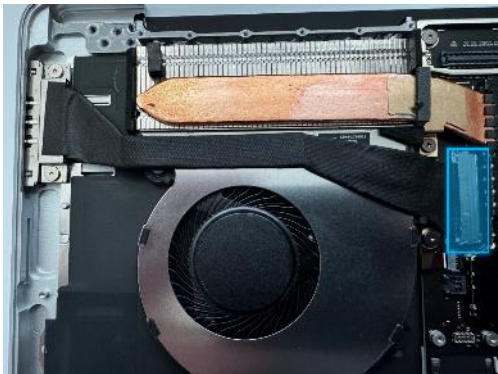
Caution: Ensure that the shield does not damage the Display FPC's during removal.

9. **Disconnect the Display Assembly FPCs** - Using a Nylon Spudger, pry up the connectors from the side and gently wiggle them free. If the connectors start giving resistance as they are being pried up, lightly push the edge being pried back down.

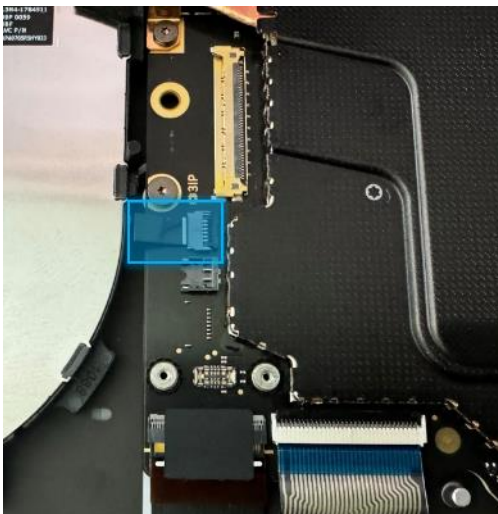
Caution: Do not force the connector if it starts to give resistance. Rock the connector back in the other direction to remove.



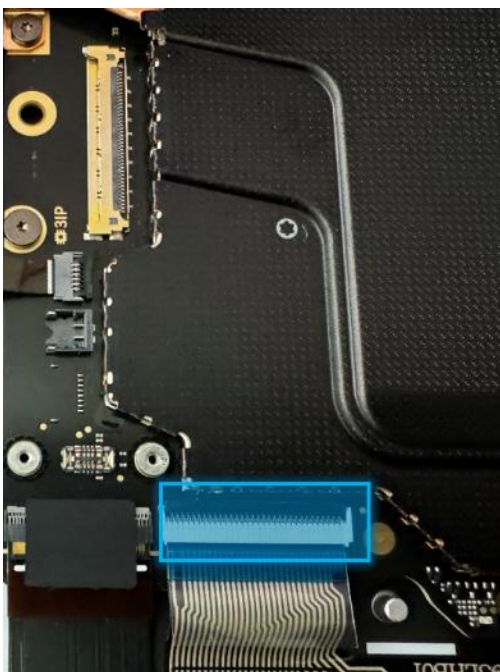
10. **Disconnect the Surface Connect Cable** – Using a Nylon Spudger, flip up the latch on the Motherboard and disconnect the Surface Connect connector from the Motherboard.



11. **Disconnect the Fan FPC** - Flip the latch on the Motherboard to disengage the lock on the Fan FPC. Remove the Fan FPC from the connector on the Motherboard.



12. **Disconnect the Keyboard FPC.** Flip the latch on the Motherboard to disengage the lock on the FPC. Remove the Keyboard FPC from the connector on the Motherboard by pulling on the black tab on the FPC.



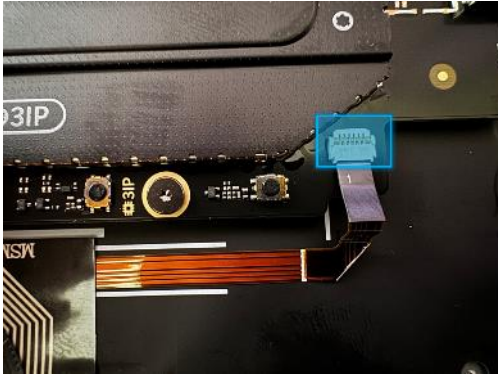
13. Disconnect the Touch FPCs –

- a. Remove the black tape covering the right Touchpad FPC and the connector. Clean the surface with IPA and cleaning swabs to ensure all residual adhesive is removed.

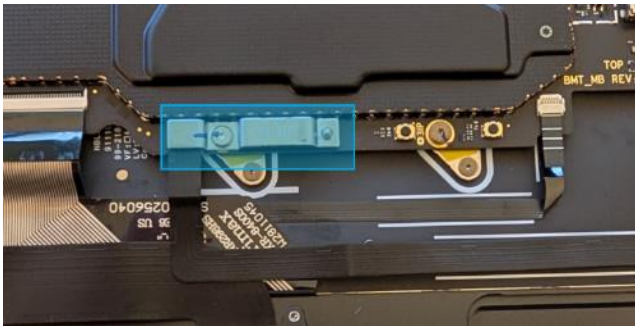


- b. Flip the latch on the Motherboard to disengage the lock on the FPC. Remove the Touchpad FPC from the connector on the Motherboard. by pulling on the tab on the FPC.

14. Disconnect the Keyboard Backlight FPC - Flip the latch on the Motherboard to disengage the lock on the FPC. Remove the Keyboard Backlight FPC from the connector on the Motherboard.



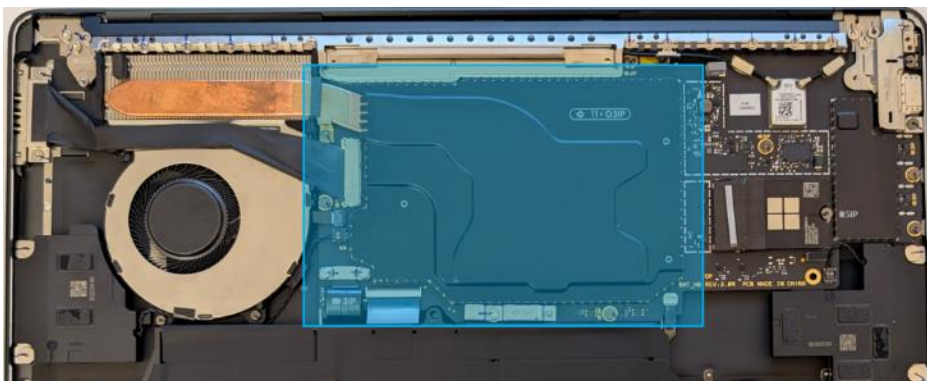
15. **Disconnect the Left Speaker** - To remove the connector from the Motherboard, pull up vertically on the wires until the connector comes free.
16. **Disconnect the Right Speaker** - To remove the connector from the Motherboard, pull up vertically on the wires until the connector comes free.
17. **Remove the Motherboard Steel Bracket** – Using a 3IP (Torx-Plus) Driver, remove the 1 screw holding the Motherboard Steel Bracket to the Enclosure. Keep the Motherboard Steel Component for re-assembly



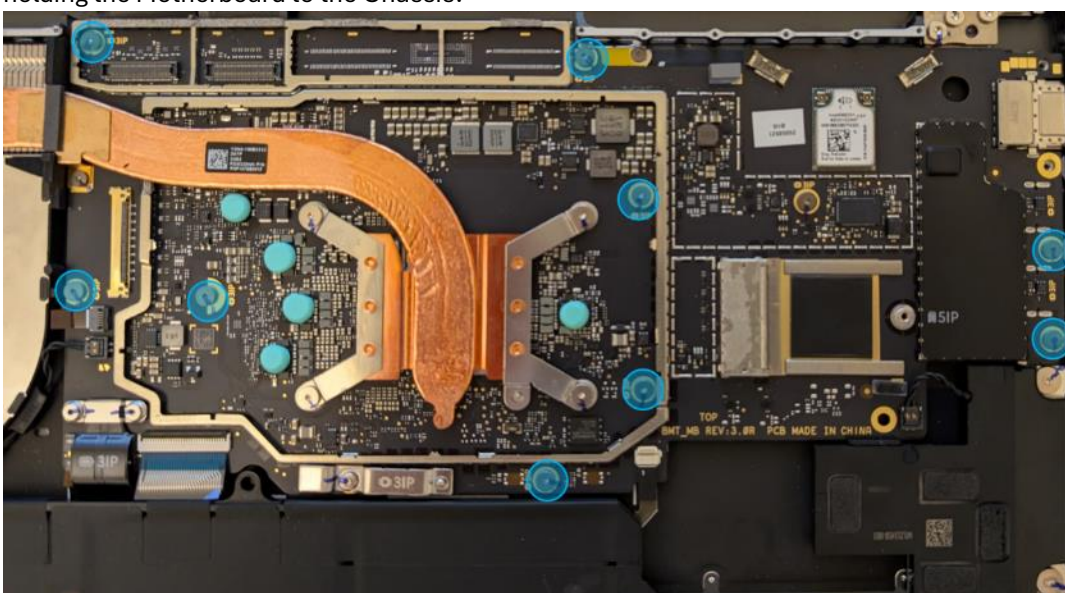
18. **15" Laptops with Smart Card Reader Only: Disconnect the Smart Card Reader FPC** - Using a nylon spudger, disconnect the Smart Card Reader FPC.



19. **Remove the T1 Shield** – Using a Nylon Spudger or ESD-Safe Tweezers, peel up the corner of the shield. Pry up the rest of the shield and remove.



20. **Remove the Motherboard Screws** - Using a 3IP (Torx-Plus) driver, remove the 10 remaining screws holding the Motherboard to the Chassis.



21. **Remove the Motherboard** – Using both hands, carefully lift the Motherboard out and up, taking care to avoid pulling on the thermal module or any connectors.
22. **15" Laptops Only: Disconnect the Micro SD Reader Cable** – Using a Nylon Spudger, flip up the latch on the Micro SD Reader board and disconnect the connector from the Motherboard.

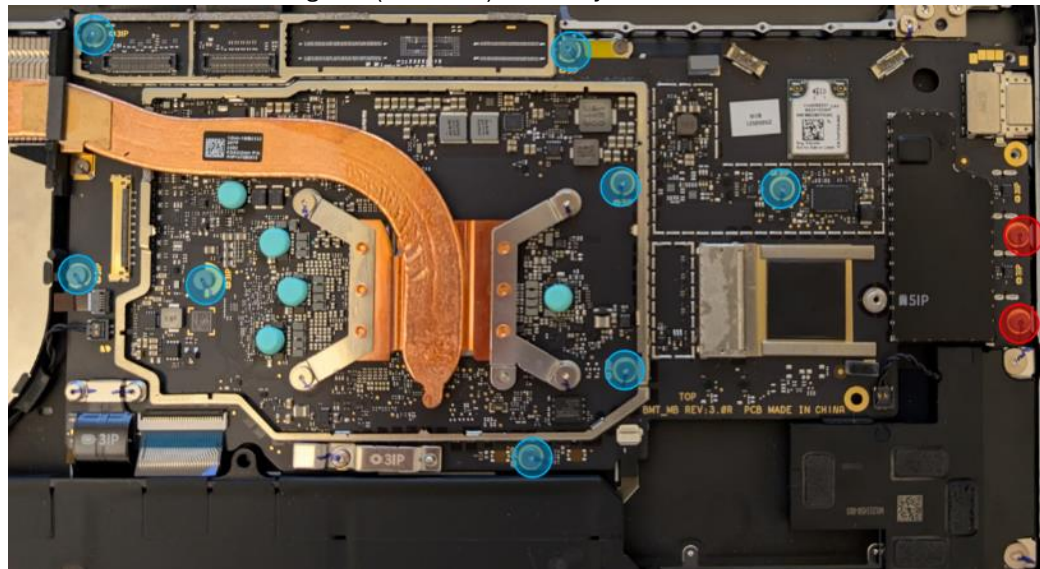


Procedure – Installation (Motherboard)

1. **15" Laptops Only: Install the Micro SD Reader Cable** – Insert the Micro SD Reader Cable into the connector on the bottom side of the Motherboard. Flip down the latch on the connector to secure it.
2. **Install the Motherboard –**

Important: The Thermal Module should be installed onto the Motherboard. Take caution during installation of the Motherboard to avoid damage to the Thermal Module. Damage to the Thermal Module will require replacement of the Motherboard or device.

- a. Lower the Motherboard USB-C side first into the device chassis. Ensure that the USB-C connectors fit in the corresponding holes in the Chassis.
- b. Lower the left side of the Motherboard and use your other hand to keep the component FPCs out of the way.
- c. Adjust the position of the Motherboard until all the holes line up with the screw bosses.
- d. Using a 3IP (Torx-Plus) driver, install 7 new Motherboard screws (**3IP - 1**) indicated in blue until just snug. Turn each screw another 45-degrees (1/8th turn) until fully fastened. Then install 2 new Motherboard screws indicated in red (**3IP - 2**), until just snug. Turn each screw another 45-degrees (1/8th turn) until fully fastened.



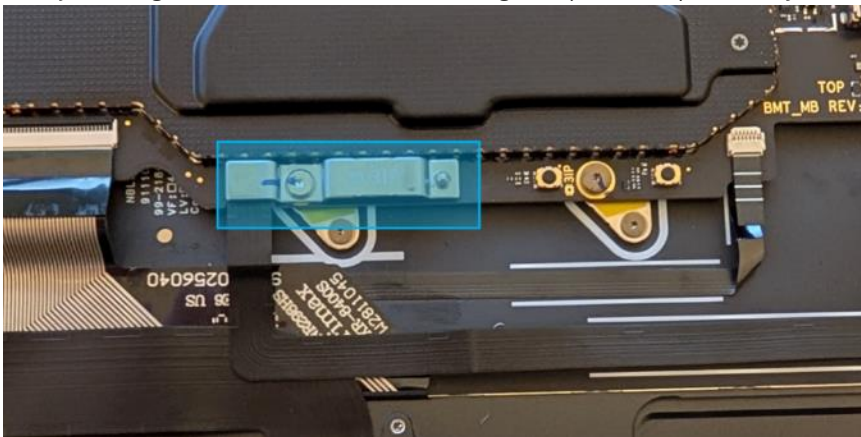
Important: Ensure that the slots next to the thermal module screws sit over the posts in the device chassis.

3. **Assemble the T1 Shield –**
 - a. Using the provided syringe of thermal paste, apply the equivalent of 1 tick mark (marked on the side of the syringe) of thermal paste to the components marked below.

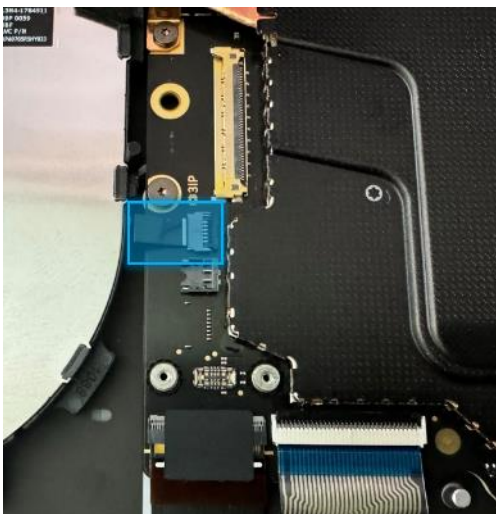


b. Assemble and install a new T1 Shield.

4. **Assemble the Motherboard Steel Bracket - For 15" Laptops with Smart Card readers, First connect the Smart Card reader FPC.** Using a 3IP (Torx-Plus) driver, install 1 new screw (🔩 3IP - 3) until just snug, and then turn another 45-degrees (1/8th turn) until fully fastened.



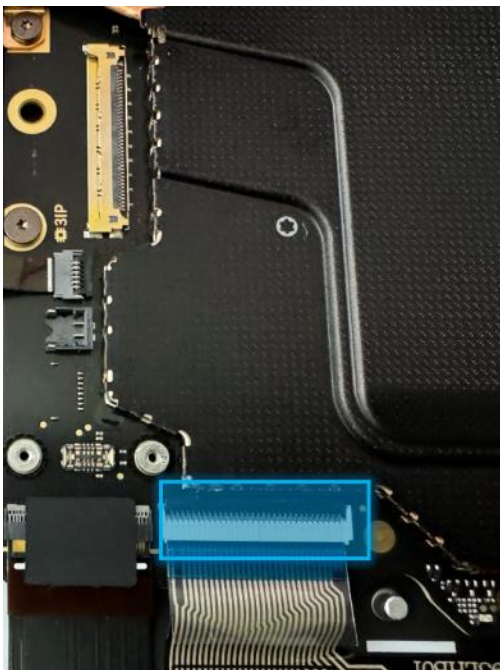
5. **Connect the Fan FPC** - Ensure the latch on the Motherboard receptacle for the Fan FPC is in a vertical position before inserting the Fan FPC. Flip the latch on the Motherboard down to secure the Fan FPC.



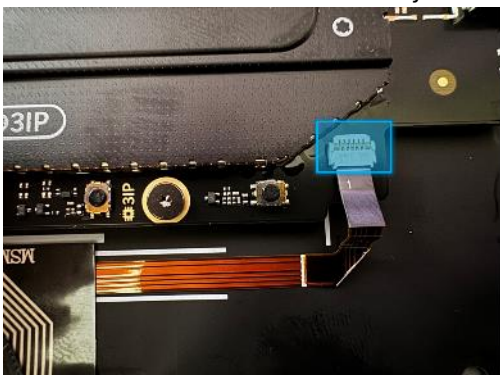
6. **Connect the Left Speaker wire** - Insert the speaker connector into the receptable on the Motherboard by pressing vertically until a snap is felt.
7. **Connect the Right Speaker wire** - Insert the speaker connector into the receptable on the Motherboard by pressing vertically until a snap is felt.
8. **Connect the Touchpad FPC -**
 - a. Ensure the latch on the Motherboard connector for the Touchpad FPC is in the vertical position before inserting the Touchpad FPC. Flip the latch on the Motherboard down to secure the Touchpad FPC.
 - b. Apply a new Touchpad FPC Tape across the FPC and the Motherboard connector.



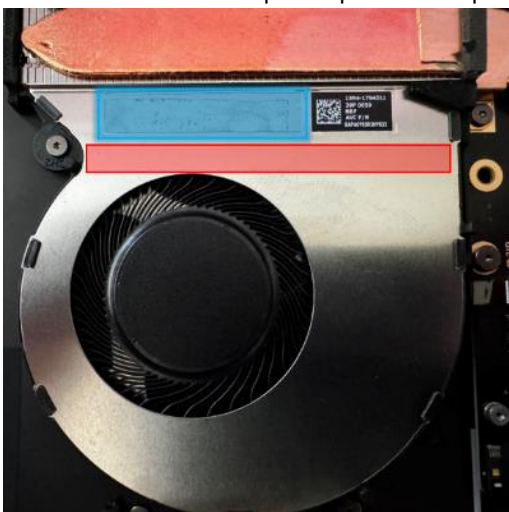
9. **Connect the Keyboard FPC** - Ensure the latch on the Motherboard connector for the Keyboard FPC is in a vertical position before inserting the Keyboard FPC. Flip the latch on the Motherboard down to secure the Keyboard FPC.



10. **Connect the Keyset Backlight FPC** - Ensure the latch on the Motherboard connector for the Keyset Backlight FPC is in a vertical position before inserting the Keyset Backlight FPC. Flip the latch on the Motherboard down to secure the Keyset Backlight FPC.



11. **Install new PSA** - Place a new piece of PSA down onto the Fan as shown. 15" Laptops with a MicroSD reader will require 2 pieces of tape. The location of the 2nd piece is indicated in red.



12. **Install the Surface Connect Cable** – Insert the connector of the Surface Connect Cable into the receptacle on the Motherboard and close the latch to secure it. There should be a click if the connector is inserted correct and the latch is fully closed.
13. **Install the Display Assembly FPC's, T3 Shield, and Antenna** - Refer to the [Procedure – Installation \(Display Assembly\)](#) section of this document for detailed instructions.
14. **Install the Audio Jack** - Refer to the [Procedure – Installation \(Audio Jack\)](#) section of this document for detailed instructions.
15. **Install the Battery** - Refer to the [Procedure – Installation \(Battery\)](#) section of this document for detailed instructions.
16. **Install the Removable Solid-State Drive** – Refer to [Procedure – Installation \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
17. **Install the Enclosure** – Refer to [Procedure – Installation \(Enclosure\)](#) section of this document for detailed instructions.

Procedure – Finalize (Motherboard)

Important: If replacing both the Motherboard Module and the Display Assembly – complete the Motherboard Module replacement prior to performing the Display Assembly Replacement to ensure proper part operation.

1. **Power on Device** – Connect a Power Supply to the device and power it on until it reaches the Windows Desktop.
2. **Connect USB** – Connect USB with the Surface Diagnostic Toolkit (SDT) loaded to an available USB port on the device under repair.
3. **Launch SDT** – From the Windows Desktop, use Windows Explorer to navigate to the USB drive. Select the SDT executable (.exe) to launch the Surface Diagnostic Toolkit.
4. **Run Touch Display Calibration** – From the SDT launch screen, select **Repair** from the drop-down menu. Next, select **Repair Setup and Validation** to enter the selection screen. Run the **Touch Display (Calibration)** tool to calibrate your new Display. Follow all on-screen instructions and allow the device to restart when prompted.

Important: If the calibration fails, reboot the device, and attempt again. If the failure continues, then the Display may be faulty and require replacement.

5. **Allow the Battery to charge** – With the device connected to a power supply, allow the battery to charge until the battery icon in Windows reads at least 50% remaining battery charge.
6. **Launch SDT** – Once the device has rebooted and is at the Windows Desktop, use Windows Explorer to navigate to the USB drive. Select the SDT executable (.exe) to launch the Surface Diagnostic Toolkit.
7. **Run Battery Authentication** – From the SDT launch screen, select **Repair** from the drop-down menu. Next, select **Repair Setup and Validation** to enter the selection screen. Select the **Battery Repair (Validation)** tool to ensure the battery is detected as properly authenticated. If the battery reads anything other than authenticated, run the Validation tool in its entirety.

Important: Battery authentication requires a stable internet connection and the latest version of the [Surface Management Extension](#). If the battery validation tool fails or is not detected properly, install the Surface Management Extension, reboot the device, and try again with a new internet connection. If failures continue, reach out to Microsoft Support.

8. **Run the Surface Diagnostic Toolkit (SDT)** – Run all diagnostics to ensure the device is functioning as expected before moving forward.

9. **Install Feet** – Refer to [Procedure – Installation \(Feet\)](#) for steps to install Feet.

Fan Replacement Process

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

Note: The procedures differ slightly between 13” and 15” Laptops. 15” Laptops will need to remove the Micro SD Reader in addition to the Surface Connect

Required Tools

- Plastic Opening Pick
- Nylon Spudger
- ESD-Safe Tweezers
- Soft ESD-Safe Mat
- 3IP (Torx-Plus) Driver
- Isopropyl alcohol (91% or greater)
- Cleaning swabs
- Anti-Static wrist strap (1M Ohm resistance)
- USB drive loaded with the Surface Diagnostic Toolkit

Primary Components

- Fan Assembly (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)
 - M1246215 Screws x 1 (Solid-State Drive) **5IP**
 - M1277572 Screws x 2 (Surface Connect) **3IP**
 - M1235134 Screws x 3 (Fan) **3IP**
 - M1167842 Tape x 1 (Right Speaker Tape)
 - M1234550 Tape x 1 (SSD)
 - M1318579 Spacer x 1 (SSD)

Additional Components (Ordered Separately)

- Feet (Refer to the Illustrated Service Parts List)

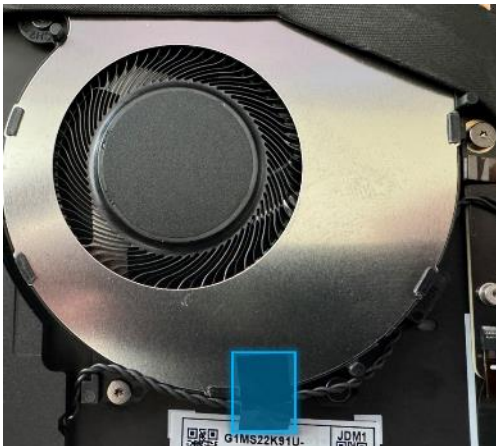
Procedure – Removal (Fan Assembly)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.
3. **Remove the Enclosure** – Refer to the [Procedure – Removal \(Enclosure\)](#) section of this document for detailed instructions.
4. **Remove the Removable Solid-State Drive** - Refer to the [Procedure – Removal \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
5. **Remove the Surface Connect** – Refer to the [Procedure – Removal \(Surface Connect\)](#) section of this document for detailed instructions.

6. **15" Laptops Only: Disconnect the Micro SD Reader Cable** – Using a Nylon Spudger, flip up the latch on the Micro SD Reader board and disconnect the connector. Peel the cable so that it is no longer adhered to the fan. Use care as to not damage the side of the cable that is connected to the Motherboard.



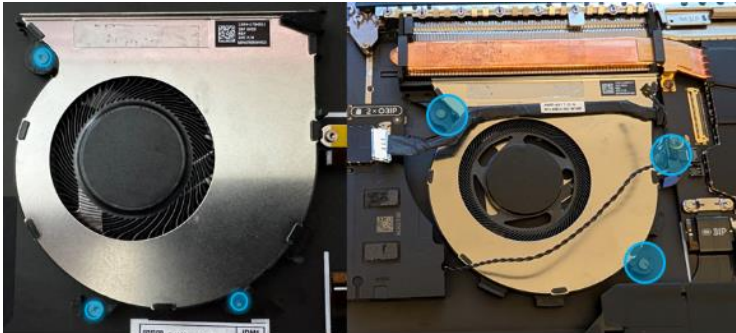
7. **13" Laptops Only: Remove the Right Speaker tape** – Remove the tape and clean the surface with IPA to remove any residual adhesive.



8. **De-route the Right Speaker Cable:** Remove the connector from the Motherboard by pulling up vertically on the wire with a nylon spudger until the connector comes free. Remove the cable from under the plastic routing guides on the fan.



8. **Remove the Fan** – Disconnect the Fan connector from the Motherboard by pulling horizontally on the protruding tab. Using a 3IP (Torx-Plus) driver, remove the 3 screws securing the Fan to the Keyboard Assembly. Note that screw locations are different on 13” and 15” Laptops. Remove the Fan from the device.



13” Laptops

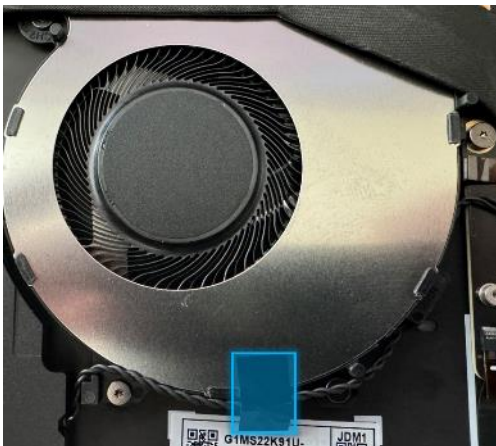
15” Laptops

Procedure – Installation (Fan)

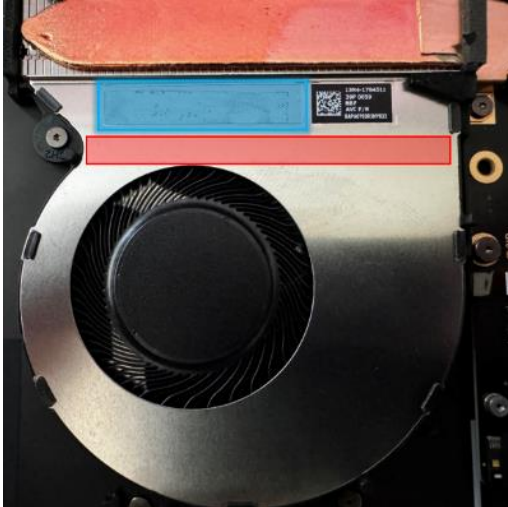
1. **Install the Fan** - Place the new Fan into the Chassis. Using a 3IP (Torx-Plus) driver, install 3 new screws. All screws should be installed until just snug, and then turned another 45-degree (1/8th turn) until fully fastened. Connect the Fan connector to the Motherboard by holding the protruding tab and inserting horizontally until the white marking is flush with the connector enclosure.
2. **Route and connect the Right Speaker wire** – Route the Right Speaker wire following the plastic routing guides on the fan. Install the speaker connector into the receptacle on the Motherboard by pressing vertically until a snap is felt.

NOTE: The connector will have 2 visible gold contacts if it is oriented in the correct direction.

3. **13” Laptops Only: Place new Speaker Tape** - Place a new Tape over the speaker wire as shown below.



4. **Install new PSA** - Place a new piece of PSA down onto the Fan as shown in blue. **15” Laptops will require 2 pieces of tape.** The location of the 2nd piece is indicated in red.



5. **15" Laptops Only: Connect the Micro SD Reader Cable** - Insert the Micro SD Reader cable into the receptacle on the Micro SD Reader board and close the latch. There should be a click if the connector is inserted correctly, and the latch is fully closed.
6. **Install the Surface Connect** - Refer to the [Procedure – Installation \(Surface Connect\)](#) section of this document for detailed instructions.
7. **Install the Removable Solid-State Drive** – Refer to [Procedure – Installation \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
8. **Install the Enclosure** – Refer to [Procedure – Installation \(Enclosure\)](#) section of this document for detailed instructions.
9. **Power on device** – Carefully place the device with the screen side facing up. Connect the device to a power supply and open the Display.
10. **Run the Surface Diagnostic Toolkit (SDT)** – Sitting at the desktop, insert the USB drive containing SDT and launch the program. Run all diagnostics to ensure the device is functioning as expected before moving forward.
11. **Power off the device** – Once the SDT tests have completed, power down the device and close the display. Invert the device so that the bottom of the device is facing up.
12. **Install the Feet** - Refer to the [Procedure – Installation \(Feet\)](#) section of this document for detailed instructions.

Smart Card Reader Replacement Process

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

Note: Only certain 15" Laptops will have a Smart Card Reader

Required Tools

- Plastic Opening Pick
- Nylon Spudger
- ESD-Safe Tweezers
- Soft ESD-Safe Mat
- 5IP (Torx-Plus) Driver
- Isopropyl alcohol (91% or greater)

- Cleaning swabs
- Anti-Static wrist strap (1M Ohm resistance)
- USB drive loaded with the Surface Diagnostic Toolkit

Primary Components

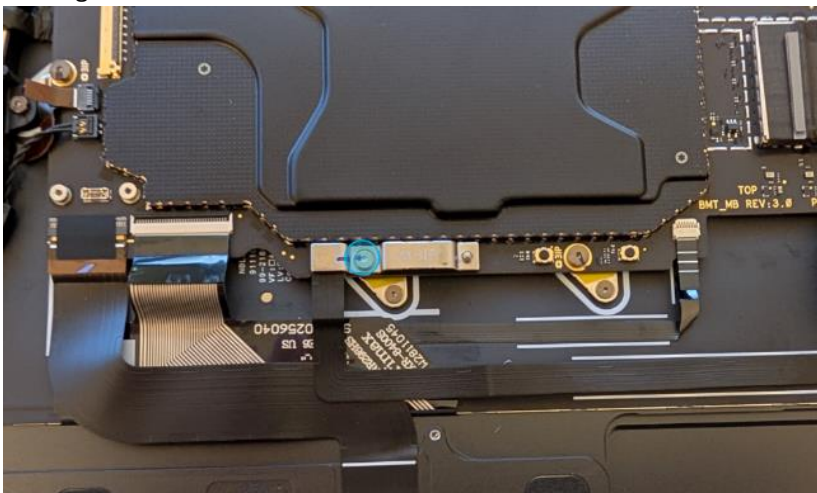
- Smart Card Reader Assembly (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)
 - M1266593 Screws x 2 (Battery FPC Bracket) **3IP - 1**
 - M1272782 Screws x 8 (Battery) **5IP - 2**
 - M1246215 Screws x 1 (Solid-State Drive) **5IP**
 - M1277573 Screws x 1 (Motherboard) **3IP - 3**
 - M1272784 Screws x 4 (Smart Card Reader)
 - M1308281 Tape x 1 (Smart Card Reader)
 - M1234550 Tape x 1 (SSD)
 - M1318579 Spacer x 1 (SSD)

Additional Components (Ordered Separately)

- Feet (Refer to the Illustrated Service Parts List)

Procedure – Removal (Keyboard Assembly)

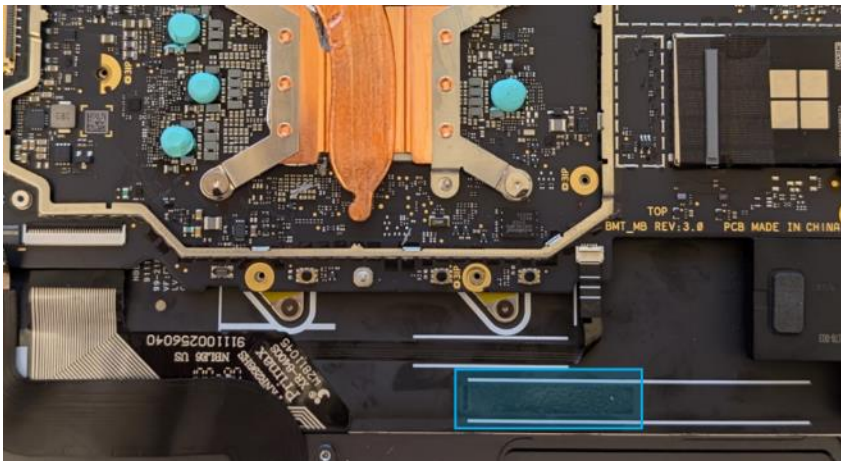
1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.
3. **Remove the Enclosure** – Refer to the [Procedure – Removal \(Enclosure\)](#) section of this document for detailed instructions.
4. **Remove the Removable Solid-State Drive** - Refer to the [Procedure – Removal \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
5. **Remove the Battery** – Refer to the [Procedure – Removal \(Battery\)](#) section of this document for detailed instructions.
6. **Remove the Smart Card Reader FPC Bracket** – Using a 3IP (Torx-Plus) driver, remove the screw holding down the Smart Card Reader FPC bracket.



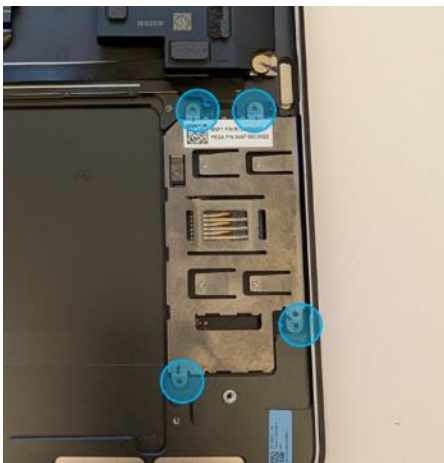
7. **Disconnect the Smart Card Reader FPC** – Using a nylon spudger, disconnect the Smart Card Reader FPC.



8. **Remove PSA** - Gently peel up the PSA (pressure-sensitive adhesive) that adheres the Smart Card Reader FPC to the Chassis and remove. Clean up the surface with IPA and cleaning swabs until all residual adhesive has been removed and the surface is clean.



9. **Remove the Smart Card Reader** - Using a 3IP (Torx-Plus) driver, remove the 4 screws securing the Smart Card Reader to the Chassis. Remove the Smart Card Reader.

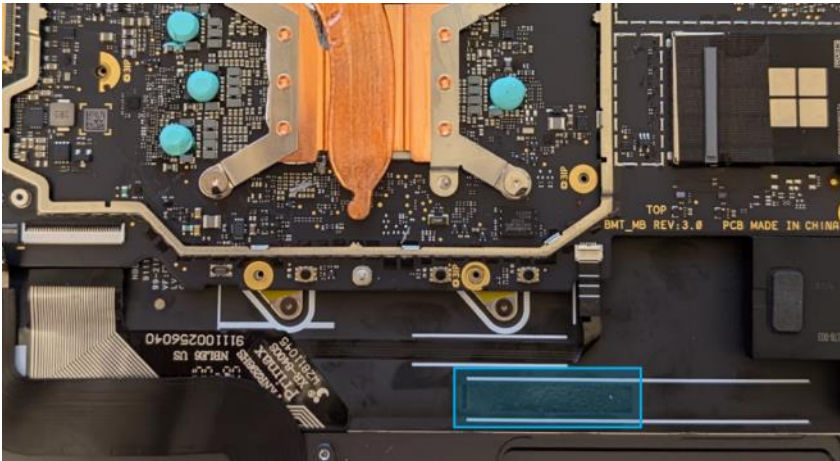


Procedure – Installation (Fan)

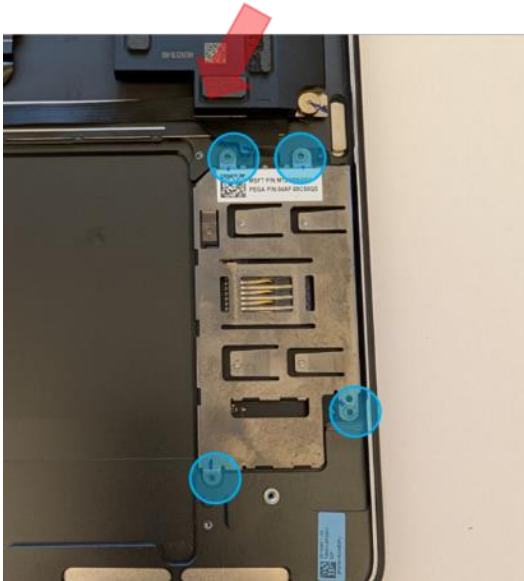
1. **Apply Smart Card Reader Tape** – Apply the Smart Card Reader Tape so that it covers the gold features on the bottom of the Smart Card Reader as shown.



2. **Install new PSA** - Place a new piece of PSA down onto the Chassis as shown.



3. **Install Smart Card Reader** – Mount the Smart Card Reader into the Chassis using a 3IP (Torx-Plus) driver with 4 screws at the locations shown. Ensure that the Smart Card Reader FPC is routed under the left speaker in the region indicated by the red arrow.



4. **Connect the Smart Card Reader FPC** – Connect the Smart Card Reader FPC.
5. **Install the Smart Card Reader FPC Bracket** – Using a 3IP (Torx-Plus) driver, install the screw holding down the Smart Card Reader FPC bracket
6. **Install the Battery** - Refer to the [Procedure – Installation \(Battery\)](#) section of this document for detailed instructions.
7. **Install the Removable Solid-State Drive** – Refer to [Procedure – Installation \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
8. **Install the Enclosure** - Refer to the [Procedure – Installation \(Enclosure\)](#) section of this document for detailed instructions.
9. **Power on device** – Carefully place the device with the screen side facing up. Connect the device to a power supply and open the Display.
10. **Run the Surface Diagnostic Toolkit (SDT)** – Sitting at the desktop, insert the USB drive containing SDT and launch the program. Run all diagnostics to ensure the device is functioning as expected before moving forward.
11. **Power off the device** – Once the SDT tests have completed, power down the device and close the display. Invert the device so that the bottom of the device is facing up.
12. **Install the Feet** - Refer to the [Procedure – Installation \(Feet\)](#) section of this document for detailed instructions.

Keyboard Replacement Process

Preliminary Requirements

Important: Be sure to follow all special (bolded) notes of caution within each process section.

Required Tools

- Plastic Opening Pick
- Nylon Spudger
- ESD-Safe Tweezers
- Soft ESD-Safe Mat
- 6IP (Torx-Plus) Driver
- 5IP (Torx-Plus) Driver
- 3IP (Torx-Plus) Driver

- 2IP (Torx-Plus) Driver
- Isopropyl alcohol (91% or greater)
- Cleaning swabs
- Anti-Static wrist strap (1M Ohm resistance)
- USB drive loaded with the Surface Diagnostic Toolkit

Primary Components

- Keyboard Assembly (Refer to the Illustrated Service Parts List)
 - M1301718 Screws x 4 (Foot screws)
 - M1266593 Screws x 2 (Battery FPC Bracket)  **3IP - 1**
 - M1243031 (13" Laptops) or M1272782 (15" Laptops) Screws x 8 (Battery)  **5IP - 2**
 - M1246215 Screws x 1 (Solid-State Drive)  **5IP**
 - M1265600 Screws x 1 (Hinge & Chassis)  **6IP**
 - M1274578 Screws x 14 (Antenna)  **3IP**
 - M1212080 Screws x 1 (Upper Audio Jack Bridge)  **3IP - 1**
 - M1330715 Screws x 1 (Lower Audio Jack Bridge)
 - M1263960 Screws x 1 (Audio Jack)  **3IP - 2**
 - M1235995 Screws x 8 (Motherboard)  **3IP - 1**
 - M1263961 Screws x 2 (Motherboard)  **3IP - 2**
 - M1277573 Screws x 1 (Motherboard)  **3IP - 3**
 - M1211014 Screws x 4 (Speakers)  **3IP**
 - M1277572 Screws x 2 (Surface Connect)  **3IP**
 - M1235134 Screws x 3 (Fan)  **3IP**
 - M1249236 Screws x 4 (13" Laptops) or x 12 (15" Laptops) (Mounting Brackets)
 - M1265416 Shield x 2 (Coax Cable Lid)
 - M1330714 Shield x 1 (T1 Shield) (13" Laptops)
 - M1330703 Shield x 1 (T1 Shield) (15" Laptops)
 - M1271924 Shield x 1 (T3 Shield)
 - M1288973 Foam x 1 (T3 Shield Foam #1)
 - M1288974 Foam x 1 (T3 Shield Foam #2)
 - M1287120 Tape x 1 (Touchpad FPC Tape)
 - M1167842 Tape x 1 (Right Speaker Tape)
 - M1328581 Tape x 1 (Display Assembly FPC Tape)
 - M1328578 Tape x 1 (Display Assembly FPC Tape)
 - M1019757 Syringe x 1 (Thermal Paste)
 - M1301902 PSA x 1 (Surface Connect & Fan)
 - M1234550 Tape x 1 (SSD)
 - M1318579 Spacer x 1 (SSD)

Additional Components (Ordered Separately)

- Feet (Refer to the Illustrated Service Parts List)

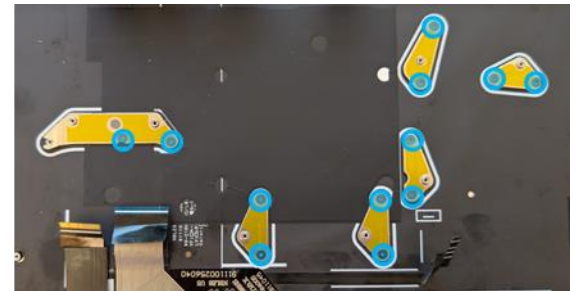
Procedure – Removal (Keyboard Assembly)

1. **Place Device** – Carefully place the closed device Display side down with the Feet facing up on a soft ESD-Safe Mat.
2. **Remove the Feet** – Refer to the [Procedure – Removal \(Feet\)](#) section of this document for detailed instructions.

3. **Remove the Enclosure** – Refer to the [Procedure – Removal \(Enclosure\)](#) section of this document for detailed instructions.
4. **Remove the Removable Solid-State Drive** - Refer to the [Procedure – Removal \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
5. **Remove the Audio Jack** - Refer to the [Procedure – Removal \(Audio Jack\)](#) section of this document for detailed instructions.
6. **Remove the Battery** – Refer to the [Procedure – Removal \(Battery\)](#) section of this document for detailed instructions.
7. **Remove the Left Speaker** – Refer to the [Procedure – Removal \(Left Speaker\)](#) section of this document for detailed instructions.
8. **Remove the Right Speaker** - Refer to the [Procedure – Removal \(Right Speaker\)](#) section of this document for detailed instructions.
9. **15" Laptops Only: Remove the Micro SD Reader** – Refer to the [Procedure – Removal \(Micro SD Reader\)](#) section of this document for detailed instructions.
10. **Remove the Display Assembly** – Refer to the [Procedure – Removal \(Display Assembly\)](#) section of this document for detailed instructions.
11. **Remove the Surface Connect** – Refer to the [Procedure – Removal \(Surface Connect\)](#) section of this document for detailed instructions.
12. **Remove the Motherboard** - Refer to the [Procedure – Removal \(Motherboard\)](#) section of this document for detailed instructions. Omit Step 22.
13. **Remove the Fan** – Refer to the [Procedure – Removal \(Fan\)](#) section of this document for detailed instructions.
14. **15" Laptops Only: Remove the Smart Card Reader** – Refer to [the Procedure – Removal \(Smart Card Reader\)](#) section of this document for detailed instructions.
15. **Remove the Mounting Brackets**– Using a 2IP (Torx-Plus) driver, remove the 2 screws holding each of the Mounting Brackets to the Keyboard Assembly. Remove the 2 (13" Laptops) or 6 (15" Laptops) Mounting Brackets from the device. Retain the Mounting Brackets.



13" Laptops



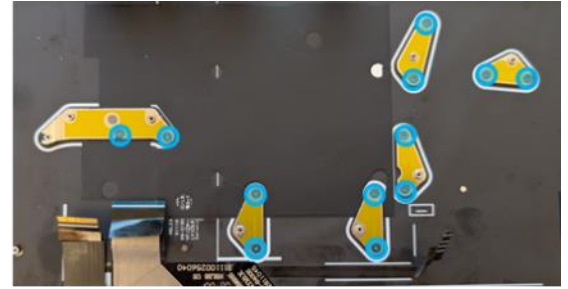
15" Laptops

Procedure – Installation (Keyboard Assembly)

1. **Install the Mounting Brackets** –
 - a. Place the 2 (13" Laptops) or 6 (15" Laptops) previously removed Mounting Brackets so that the yellow side is visible, and they are arranged as shown.



13" Laptops



15" Laptops

- b. Using a 2IP (Torx-Plus) driver, install 4 new Mounting Bracket screws until the screws are just snug, and then turn each another 45-degrees (1/8th turn) until fully fastened.
2. **15" Laptops Only: Install the Micro SD Reader** – Refer to [the Procedure – Installation \(Micro SD Reader\)](#) section of this document for detailed instructions.
3. **Install the Fan** - Refer to [Procedure – Installation \(Fan\)](#) section of this document for detailed instructions.
4. **Install the Motherboard** - Refer to the [Procedure – Installation \(Motherboard\)](#) section of this document for detailed instructions. Complete steps 1-5 in that section. Ensure the FPCs from the Keyboard Module are placed over the Motherboard.
5. **15" Laptops with Smart Card Reader Only - Install the Smart Card Reader** – Refer to [the Procedure – Installation \(Smart Card Reader\)](#) section of this document for detailed instructions.
6. **Install the Left Speaker** - Refer to [Procedure – Installation \(Left Speaker\)](#) section of this document for detailed instructions.
7. **Install the Right Speaker** - Refer to [Procedure – Installation \(Right Speaker\)](#) section of this document for detailed instructions.
8. **Continue with Motherboard installation** - Refer to the [Procedure – Installation \(Motherboard\)](#) section of this document for detailed instructions. You will start at Step 6.
9. **Install the Display** - Refer to the [Procedure – Installation \(Display\)](#) section of this document for detailed instructions.
10. **Install the Audio Jack** - Refer to the [Procedure – Installation \(Audio Jack\)](#) section of this document for detailed instructions.
11. **Install the Surface Connect** - Refer to the [Procedure – Installation \(Surface Connect\)](#) section of this document for detailed instructions.
12. **Install the Battery** - Refer to the [Procedure – Installation \(Battery\)](#) section of this document for detailed instructions.
13. **Install the Removable Solid-State Drive** – Refer to [Procedure – Installation \(Removable Solid-State Drive\)](#) section of this document for detailed instructions.
14. **Install the Enclosure** - Refer to the [Procedure – Installation \(Enclosure\)](#) section of this document for detailed instructions.
15. **Power on device** – Carefully place the device with the screen side facing up. Connect the device to a power supply and open the Display.
16. **Run the Surface Diagnostic Toolkit (SDT)** – Sitting at the desktop, insert the USB drive containing SDT and launch the program. Run all diagnostics to ensure the device is functioning as expected before moving forward.
17. **Power off the device** – Once the SDT tests have completed, power down the device and close the display. Invert the device so that the bottom of the device is facing up.

18. **Install the Feet** - Refer to the [Procedure – Installation \(Feet\)](#) section of this document for detailed instructions.

Environmental Compliance Requirements

All waste electrical and electronic equipment (WEEE), waste electronic components, waste batteries, and electronic waste residuals must be managed according to applicable laws and regulations. and H09117, “Conformance Standards for Environmentally Sound Management of Waste Electrical and Electronic Equipment (WEEE)” which is available at this link: <https://www.microsoft.com/en-pk/download/details.aspx?id=11691> . In case of questions, please contact AskECT@microsoft.com .